



STRENGTHENING HEALTH SYSTEMS AND RMNCAH-N OUTCOMES THROUGH RAPID CYCLE ANALYTICS AND DATA USE

Country Workshop: Introduction to FASTR RMNCAH-N Service Use Monitoring

Lusaka, Zambia

January 27-30, 2026

Welcome and Opening Remarks



Introductions



Workshop Objectives

The objectives of the workshop are to:

- **Introduce the FASTR HMIS analysis platform** – Familiarize participants with the platform's purpose, core features, and functionalities, including workflows, reporting options, and outputs
- **Strengthen analytical and reporting skills** – Enhance participants' capacity to manage data structures, customize visualizations, and produce tailored reports for diverse audiences
- **Prepare MoH for participation in the multi-country workshop** – Configure the FASTR analytics platform for Zambia; Produce the first quarterly report
- **Adapt the FASTR phone survey questionnaire to the Zambian context**

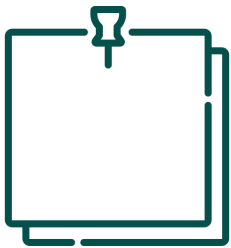
Expectations for the Workshop

Write on a sticky note and stick it on the flip chart:

What do you want to learn?

What questions about FASTR do you want answered?

We will review these on day four to see if we met our expectations.



Expected Outputs

Disruption Analysis Activities

- Data extraction and configuration of analytics platform
- Produce first quarterly report
- Review results and contextualize findings (with relevant program teams)
- Roadmap for participation in Abuja workshop

Phone Survey Activities

- Adaptation of questionnaire (with program input)

Agenda

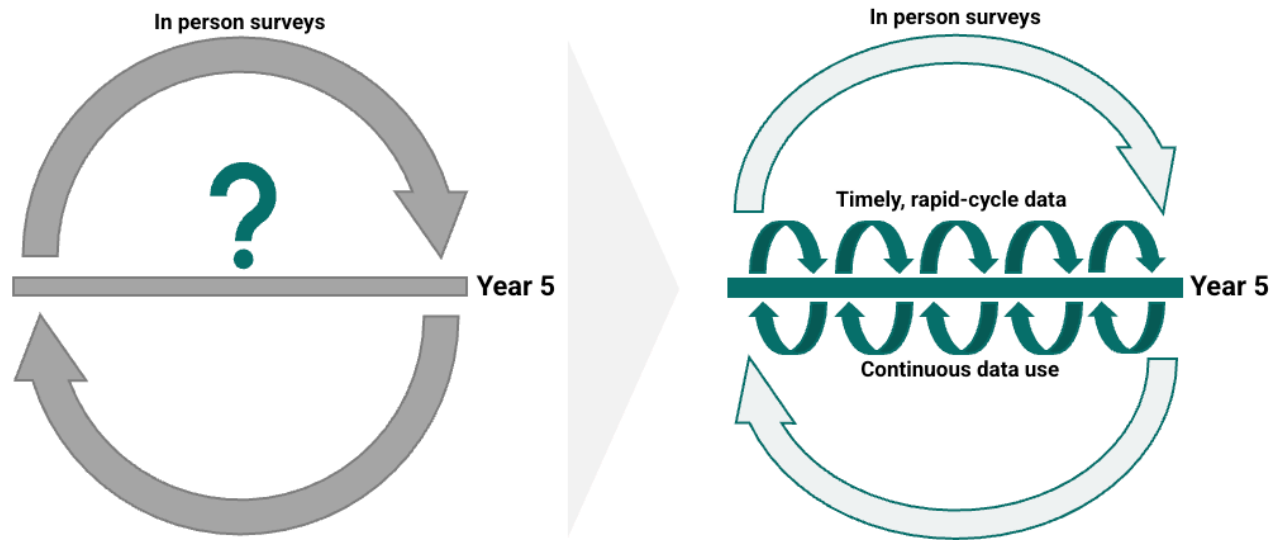
Day 1 -- Laying the Foundation: Introducing FASTR and Configuring the Analytics Platform

Time	Agenda	Facilitator/Presenter
Opening Session		
08:30–09:00	Participant registration	MoH team
09:00–09:10	Welcome and opening remarks	MoH team
09:10–09:20	Icebreakers/Introductions	MoH team
09:20–09:35	Workshop objectives and agenda	GFF FASTR team
Session 1: Overview of the FASTR approach		
09:35–10:15	Overview: FASTR Approaches	GFF FASTR team
Session 2: Identifying priority questions and indicators		
10:15–11:00	Identifying priority questions and indicators	GFF FASTR team
Session 3: HMIS data extraction		
11:00–12:00	Data extraction: Rationale and methods	GFF FASTR team
Session 4: Getting into the platform		
12:00–12:30	Getting participants into the platform	GFF FASTR team
12:30–14:00	<i>Lunch Break</i>	
Session 5: Configuring the FASTR analytics platform		
14:00–15:30	Configuring the analysis platform	GFF FASTR team
Closing		
15:30–16:00	Key messages and wrap-up	GFF FASTR team
16:00–16:30	Reflections from participants	Participants

Session 1: Overview of the FASTR approach

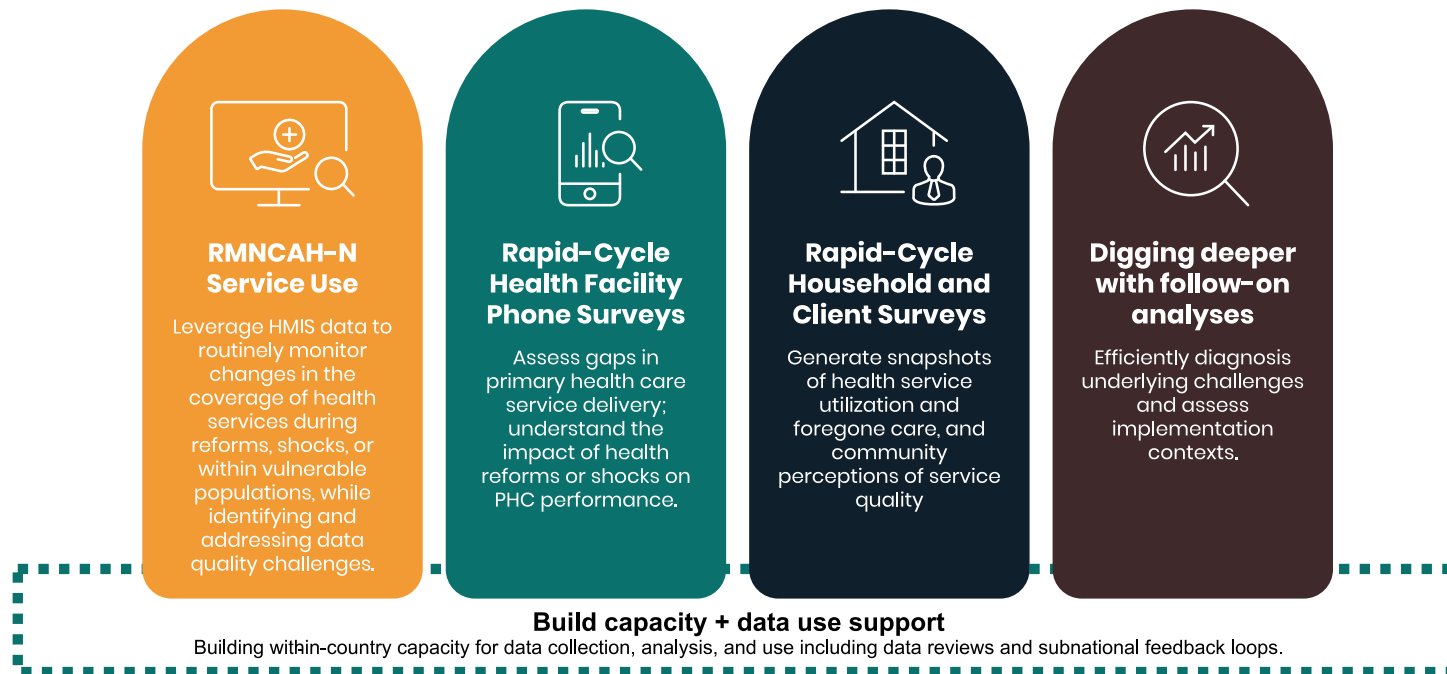
What are we trying to achieve?

Rapid cycle analytics accelerates improvements in RMNCAH-N outcomes by increasing the systematic use of data for decision making



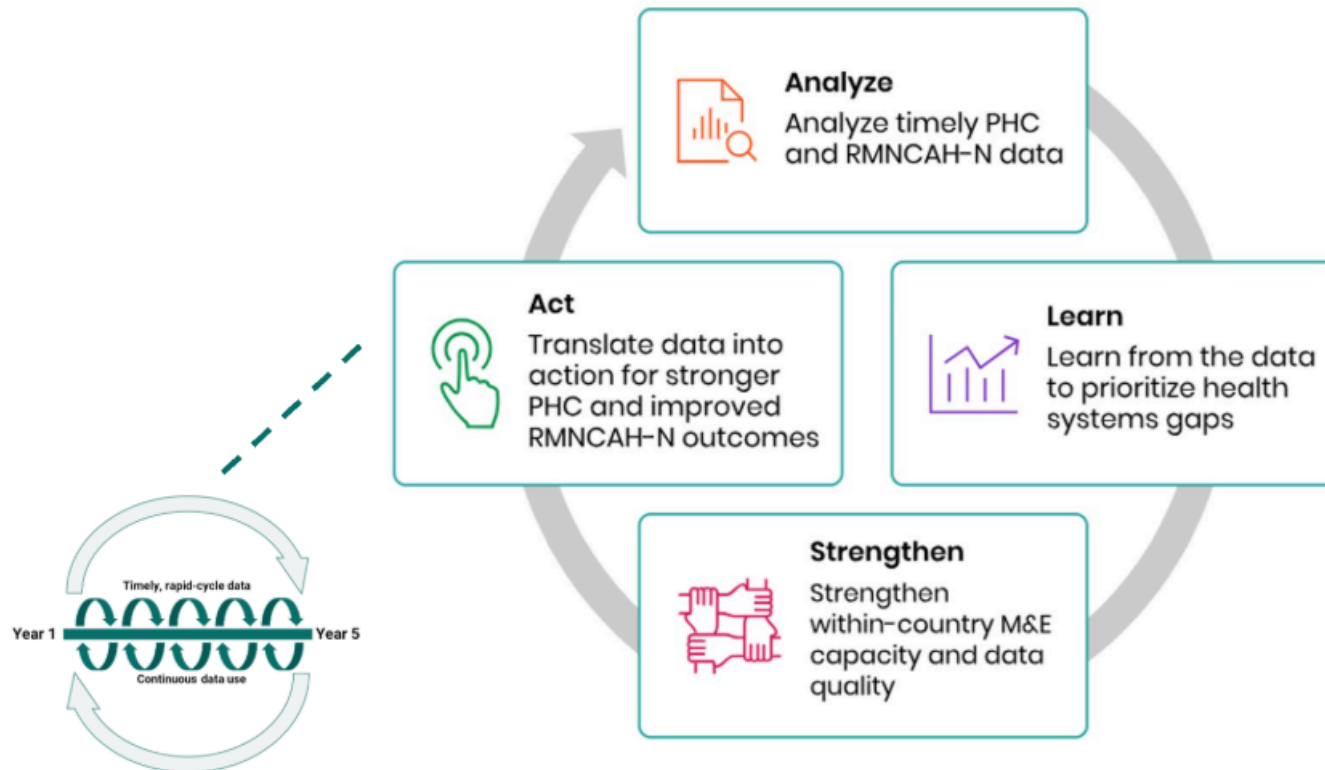
How can this be achieved?

Timely, rigorous, and low-cost approaches to monitoring PHC systems, underpinned by capacity building and data use support aligned with country demand and needs



What is FASTR?

An approach to catalyzing continuous 'analyze, learn, strengthen, act' cycles to drive the systematic use of timely data for decision making.

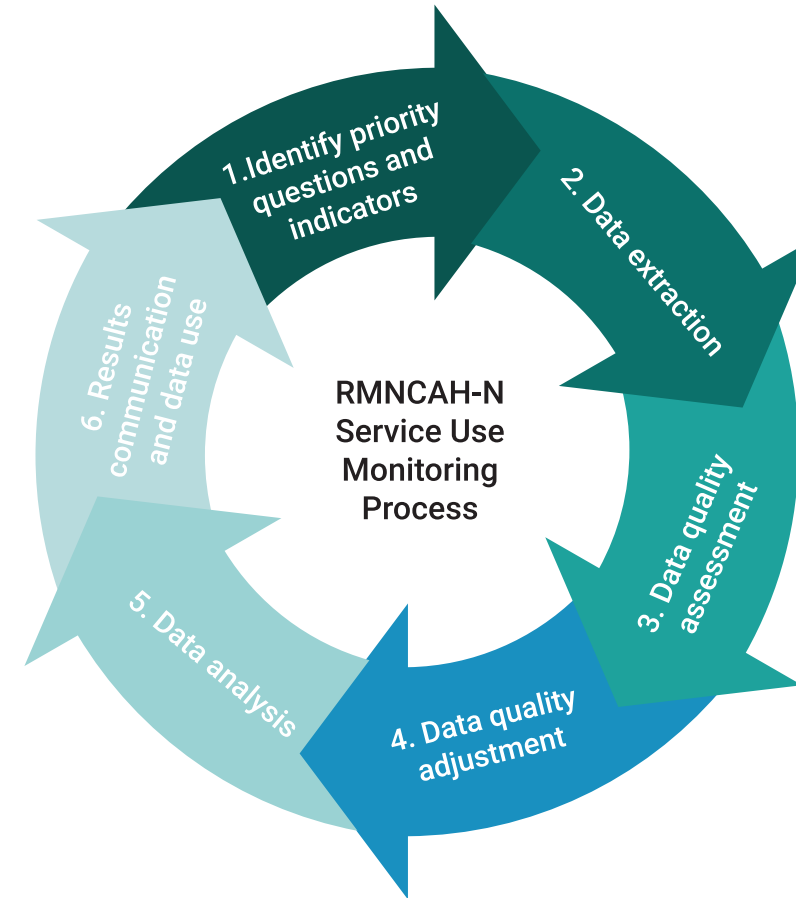


What is the FASTR approach to RMNCAH-N service use monitoring?

Quarterly analyses of DHIS2 data, focusing on prioritized national indicators

Building sustainable tools to ensure that stakeholders who need to use data can generate the right analysis and visualizations, at the right time, on their indicators of interest

Combining analysis and visualization with capacity strengthening and data use support for sustainability and institutionalization



Demo – FASTR analysis platform



Session 2: Identifying priority questions and indicators

What is a data use case?

A data use case is a specific scenario where data is utilized to achieve a particular goal or solve a problem.

Why is defining a data use case important?

- Guides decision making by providing a clear framework for analysis
- Enhances efficiency by focusing analyses on a set of relevant key indicators to solve a specific data need
- Leads to better results by aligning data efforts with organizational goals

Data use case: Application 1

Placeholder – content to be added

Data use case: Application 2

Placeholder – content to be added

Data use case: Application 3

Placeholder – content to be added

What is our common data use case that will be the focus of this workshop?

Following large shifts in resource availability from external sources, many countries are experiencing abrupt and dramatic reductions in financing

- Resulting in critical gaps in programs and systems
- Leading to potentially severe effects on service delivery and health outcomes for women, children and adolescents

Key questions arising:

- What is the magnitude of the cuts, and what effect are they having on service delivery?
- What is the optimal way to prioritize remaining resources?
- What other adaptations can safeguard and strengthen essential service delivery for women, children and adolescents?

How do we select indicators? What makes a good indicator for the FASTR analysis?

Indicator selection is critical to the quality and usefulness of FASTR analysis. Indicators should be chosen based on the following criteria:

- **Relevance** – Does this indicator provide data that aligns with our priority questions and objectives?
- **Volume** – Is this indicator collected at a high volume, which improves the robustness of analysis?
- **Completeness** – Does the indicator have a high completeness rate across reporting facilities?
- **Frequency** – Is the indicator reported frequently enough (e.g., monthly) to support rapid-cycle analysis?
- **Type** – Is this indicator a count of services delivered?

FASTR core indicators

The FASTR approach focuses on a core set of RMNCAH-N indicators that represent key points along the reproductive, maternal, newborn, child, and adolescent health and nutrition continuum in low- and middle-income countries.

These indicators typically have higher reporting volumes and completeness and serve as proxies for broader service delivery patterns.

- Antenatal client 1st visit
- Antenatal client 4th visit
- Institutional delivery
- Postnatal care 1
- BCG doses
- Pentavalent 1st dose
- Pentavalent 3rd dose
- Outpatient visits

Why focus on high-volume indicators?

One of the core strengths of the FASTR approach is its ability to adjust for data quality issues. High-volume indicators are better suited to this process because:

Reduced sensitivity to outliers

In low-volume indicators, individual data points can disproportionately affect trends.

More stable estimates

High-volume data reduce random variability and improve the reliability of trend detection.

Clearer identification of true anomalies

Larger counts make it easier to distinguish genuine outliers from natural variation.

Why focus on high-completeness indicators?

Indicators with high reporting completeness are preferred because they:

Improve data reliability

More complete data reduces bias and provide a more representative picture of service delivery.

Support consistent analysis

High completeness enables meaningful comparisons across time and geographic areas.

Reduce misinterpretation

Incomplete data can falsely suggest changes in service utilization when changes are driven by reporting gaps rather than real trends.

While statistical methods such as imputation can be used to address incomplete data, these methods require assumptions about missing values.

Why focus on count indicators?

Limitations of proportion indicators

- Proportions limit the ability to adjust numerators and denominators separately for data quality issues
- Numerators and denominators may each be affected by different sources of error
- Separating counts from denominator estimation allows for more transparent and flexible adjustment

Mortality as a rare event

- Mortality indicators are typically low-frequency and not well suited to adjustment
- These indicators are generally better analyzed using annual rather than monthly or quarterly data

Countries have selected indicators to align with the disruptions context and country priorities

		Indicator mapping for RMNCAH-N Service Use Monitoring using Administrative Data				
Data Source Mapping				Indicator Definition		
INDICATOR CATEGORY ▼	INDICATOR OF INTEREST ▼	INDICATOR SOURCE (national strategy, WB project, FASTR core indicators, etc) ▼	IS THIS INDICATOR IN DHIS2? ▼	OFFICIAL INDICATOR NAME IN DHIS2 ▼	INDICATOR JSON ID ▼	NOTES ▼
RMNCAH+N	ANC1	FASTR core indicators				
RMNCAH+N	ANC4	FASTR core indicators				
RMNCAH+N	Deliveries – institutional	FASTR core indicators				
RMNCAH+N	BCG	FASTR core indicators				
RMNCAH+N	PNC1	FASTR core indicators				
RMNCAH+N	Penta1	FASTR core indicators				
RMNCAH+N	Penta3	FASTR core indicators				
RMNCAH+N	OPD visits	FASTR core indicators				

The FASTR Data Prep Checklist has been shared with countries.

The checklist includes the FASTR core RMNCAH-N indicators.

Countries have added additional indicators that may be particularly relevant to the country context (e.g., indicators related to services expected to be impacted by recent funding shifts, priority indicators for the government and/or the WB project).

These are the indicators included in the current analysis. Countries can continue to add indicators over time as needed for their use case(s).

Activity



We are going to review the Zambia data prep checklist together and discuss for each indicator:

Relevance

Does this indicator provide data that aligns with our priority questions and objectives?

Volume

Is this indicator collected at a high volume, which improves the robustness of analysis?

Completeness

Does the indicator have a high completeness rate across reporting facilities?

Frequency

Is the indicator reported frequently enough (e.g., monthly) to support rapid-cycle analysis?

Type

Is this indicator a count of services delivered?

Based on the discussion we can refine the selection of indicators.

Session 3: HMIS data extraction

Show of hands...



Do you regularly extract data from DHIS2?

If so, what are the primary reasons?

Why would you extract data from DHIS2? Why not just do analysis in DHIS2 itself?

Data quality adjustment

The FASTR approach focuses on data quality adjustments to expand the analyses countries can do with DHIS2 data and to generate more robust estimates.

Analysis complexity

The FASTR approach uses more advanced statistical methods, such as regression analysis, which are not available in DHIS2. While DHIS2 can plot trends over time using raw data, FASTR can go further by identifying significant increases or decreases in service volume, adjusting for data quality issues, accounting for expected seasonal variations, and comparing key periods, such as before and after a reform.

The choice between DHIS2 and the FASTR approach should be guided by the specific purpose of your analysis. Select the tool that best aligns with your analytical needs!

Data format and granularity

Data format example

Data should be downloaded for each indicator of interest, at facility level, and monthly for the period of interest.

Data should be saved in long format meaning each row represents a single observation or measurement (see example).

Data should be saved in .csv format and can be saved in either a single .csv file or multiple .csv files which will be combined when uploading to the analysis platform.

How much data?

Initial FASTR analysis

- Generally recommended to download approximately five years of historical data
- However, the exact period should be determined based on data availability, consistency in indicator definitions over time, and the specifics of a country's routine data system
- Ideally, using at least five years of historical data allows for a thorough assessment of trends over time

Routine update to FASTR analysis

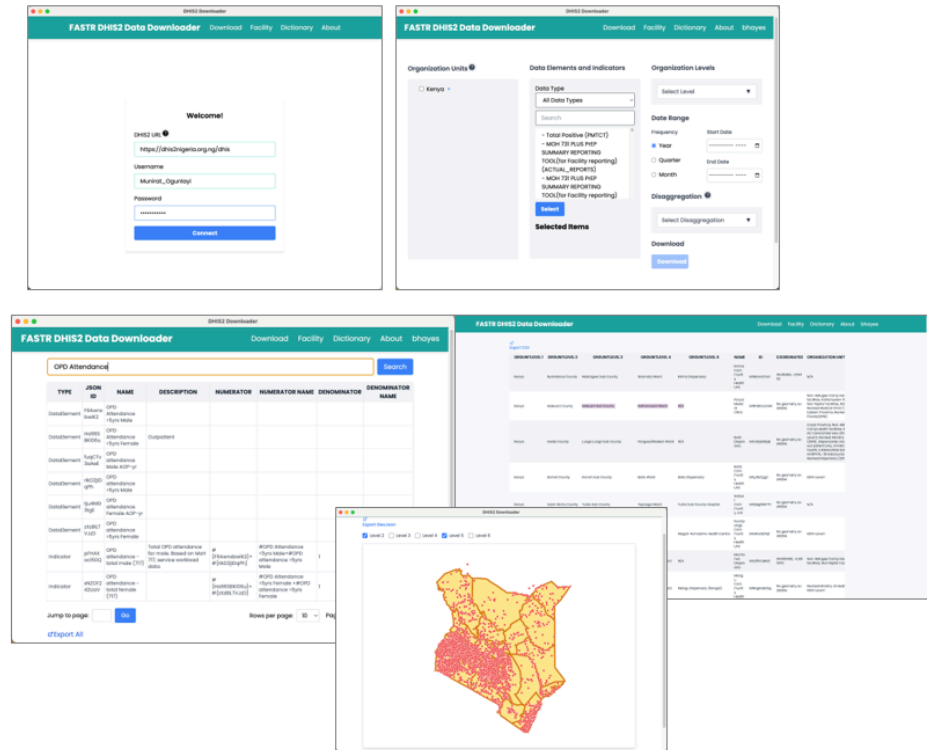
- Start with the existing database and download new data covering the most recent months not previously included – this is usually a three-month period when the FASTR analysis is being implemented on a quarterly basis
- Additionally, include the three preceding months to the new data time period, as this relatively recent data is often subject to changes due to late reporting or data quality adjustments
- If you have reason to believe there have been substantial changes to the historical data, you can always choose to redownload a longer time period

Data extraction

We offer two tools for bulk DHIS2 data extraction: a user-friendly Data Downloader and a direct import feature within the FASTR analytics platform.

The Data Downloader provides a streamlined interface to download DHIS2 data. This tool is particularly useful to explore DHIS2 metadata and download indicators requiring disaggregated dimensions.

The Data Downloader is available at:
<https://github.com/worldbank/DHIS2-Downloader/releases/>



DHIS2 Data Downloader

The Data Downloader is a desktop application for extracting data from DHIS2.

Key features:

- Connect to any DHIS2 instance
- Browse and select data elements and indicators
- Download facility-level data in CSV format
- Maintain download history

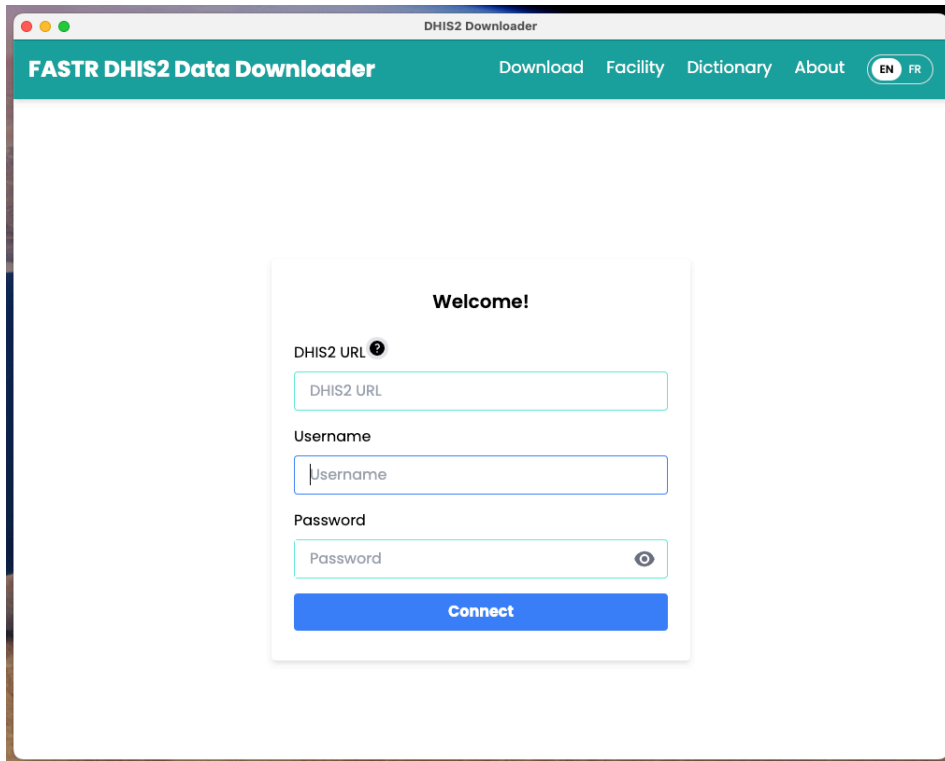
Download from GitHub:

<https://github.com/worldbank/DHIS2-Downloader/releases/>



Facilitator will demonstrate the Data Downloader

Data Downloader: Login



The screenshot shows a web application window titled "DHIS2 Downloader". The header bar is teal and contains the text "FASTR DHIS2 Data Downloader" on the left, and navigation links "Download", "Facility", "Dictionary", "About", and language toggles "EN" and "FR" on the right. The main content area is white and features a central login form. The form is titled "Welcome!" and contains three input fields: "DHIS2 URL" with a help icon, "Username", and "Password" with a toggle icon. A blue "Connect" button is positioned at the bottom of the form.

Connect to your DHIS2 instance

- Enter your DHIS2 server URL
- Provide your username and password
- The tool securely stores credentials for future sessions

Data Downloader: Overview

Main interface

- Browse available data elements and indicators
- Select time periods and organization units
- Configure download options
- Start data extraction

FASTR DHIS2 Data Downloader [Download](#) [Facility](#) [Dictionary](#) [About](#) [worldbank](#)

Organization Units [?] **1**

☐ Sierra Leone +

Data Elements and Indicators

Data Type

All Data Types **2**

Search

- hsp_Consumables_Rapid test kit
Hepatitis C (HCV), pcs_L/A_Comments
- MFRHC_hsp_Laboratory
(ACTUAL_REPORTS)
- MFRHC_hsp_Laboratory
(ACTUAL_REPORTS_ON_TIME)
- MFRHC_hsp_Laboratory
(EXPECTED_REPORTS)

Select

Selected Items

Organization Levels **3**

Select Level

Date Range

Frequency **4**

Start Date **5**

End Date

Disaggregation [?] **6**

Select Disaggregation

Download

Note: Use a unique file name to prevent overwriting.

Download **7**

Data Downloader: Download history

Track your downloads

- View all previous download sessions
- Re-download data with same parameters
- Access download logs and status
- Manage downloaded files

FASTR DHIS2 Data Downloader [Download](#) [Facility](#) [Dictionary](#) [About](#) [worldbank](#)

[Export History](#) [Delete](#) [Re-download](#) **5**

4	☐	ID	NOTES	ORGANIZATIONLEVEL	PERIOD	DIMENSION	DIMENSIONNAME	DISAGGREGATION	URL
☐	Edit Save Pass	1		LEVEL-5	202201:202202:202203	grudkdaw9h	- Antenatal client 4th visit		https://sl.dhis2.org/hmis/api/analytics.csv?dimension=ou1LEVEL-5&dimension=pe202201:202202:202203&dimension=dxgrudkdaw9h&displayProperty=NAME&ignoreLimit=TRUE&hierarchyMeta=true&hideEmptyRows=TRUE&showHierarchy=true&rows=ou.pe.dx

2 **3** **1** Jump to page: [Go](#) Rows per page: Page 1 of 1 [<](#) [>](#)

Data Downloader: Data dictionary

FASTR DHIS2 Data Downloader

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1antenatal client 1st

Search

TYPE	JSON ID	NAME	DESCRIPTION	NUMERATOR	NUMERATOR NAME	DENOMINATOR	DENOMINATOR NAME
DataElement	A9ui8RnoATt	Antenatal client 1st visit haemoglobin done					
DataElement	CqD66BAZA5z	Antenatal client 1st visit screened for syphilis					
DataElement	DUCOdGZXNHQ	NACP-PMTCT - Antenatal client 1st visit (from HF3 form)					
DataElement	Hfej6DufGQ	Antenatal client 1st promotional visit					
DataElement	pdNVqI95gs	Antenatal client 1st visit under 12 weeks					
DataElement	yl2yw27ekfR	Antenatal client 1st visit					
Indicator	Deom56AwJrZ	Antenatal client 1st visit who had haemoglobin test (%)_K	Pregnant women who had their haemoglobin test during their ANC 1st visit	# {A9ui8RnoATt}	#Antenatal client 1st visit haemoglobin done	# {yl2yw27ekfR}	#Antenatal client 1st visit
Indicator	FxjXWfv3HL4	Antenatal client 1st visit under 12 weeks rate_X	proportion of ANC 1st visits that occurred under 12 weeks	# {pdNVqI95gs}	#Antenatal client 1st visit under 12 weeks	# {yl2yw27ekfR}	#Antenatal client 1st visit

Jump to page: Go

Rows per page: 10 Page 1 of 1 < >

Export All

3

Export This Page

4

Explore available data

- Browse all data elements from your DHIS2
- Search by name or code
- View metadata and definitions
- Identify indicators for your analysis

Data Downloader: Facility list

FASTR DHIS2 Data Downloader

[Download](#)[Facility](#)[Dictionary](#)[About](#)[worldbank](#)

[Export CSV](#)

LEVEL	ORGUNITLEVEL 1	ORGUNITLEVEL 2	ORGUNITLEVEL 3	ORGUNITLEVEL 4	NAME	ID	COORDINATES	ORGANIZATION UNIT GROUPS
5	Sierra Leone	Kono District	Kono District Council	Kamara Chiefdom	Kamara Commu nities	A0UqOtjW8Of	-11.026827, 8.721628	Communities (Placeholders), Public
5	Sierra Leone	Kailahun District	Kailahun District Council	Luawa Chiefdom	Bailu Port of Entry	A10JHz59FN2	No geometry available	Port of Entry, Public
5	Sierra Leone	Pujehun District	Pujehun District Council	Gallinest Chiefdom	Kpowubu MCHP	A1OP0onyJIX	-11.5709, 7.342915	MCHP, PMTCT Facilities, PHUs, eIDSR-Reporting facilities, Public, Implant
5	Sierra Leone	Koinadugu District	Koinadugu District Council	Kamukeh Chiefdom	Thellia Port of Entry	A1pv9kyBwU4	-11.7989, 9.9986	Port of Entry, Public
5	Sierra Leone	Falaba District	Falaba District Council	Kamadugu Yirala Chiefdom	Dankawalle CHC	A1z9IPUnnJp	-11.331542, 9.629802	CHC, PHUs, eIDSR-Reporting facilities, Public, Implant
5	Sierra Leone	Pujehun District	Pujehun District Council	Peje Chiefdom	Pejewara (Futa Peje) MCHP	A6dVJUNJLU	-11.506295, 7.587257	MCHP, PMTCT Facilities, PHUs, eIDSR-Reporting facilities, Public, Implant
5	Sierra Leone	Kailahun District	Kailahun District Council	Kissi Kama Chiefdom	Sangha Port of Entry	A8lty0wW9c	-10.415372, 8.436378	Port of Entry, Public
5	Sierra Leone	Kenema District	Kenema City Council	Kenema City	Lango Town MCHP	A82aB5gRlth	-11.17263, 7.860986	MCHP, PMTCT Facilities, PHUs, eIDSR-Reporting facilities, Public, Implant
5	Sierra Leone	Bonthe District	Bonthe District Council	Sittia Chiefdom	Mbakie MCHP	A9MVWMOILZh	-12.6161, 7.6369	MCHP, PHUs, eIDSR-Reporting facilities, Public
5	Sierra Leone	Western Area Urban District	Freetown City Council	East 3 Zone	Egyptia n (Calaba Town) Clinic	A9zxwZDuOHP	-13.17487, 8.46455	eIDSR-Reporting facilities, Clinic, Private

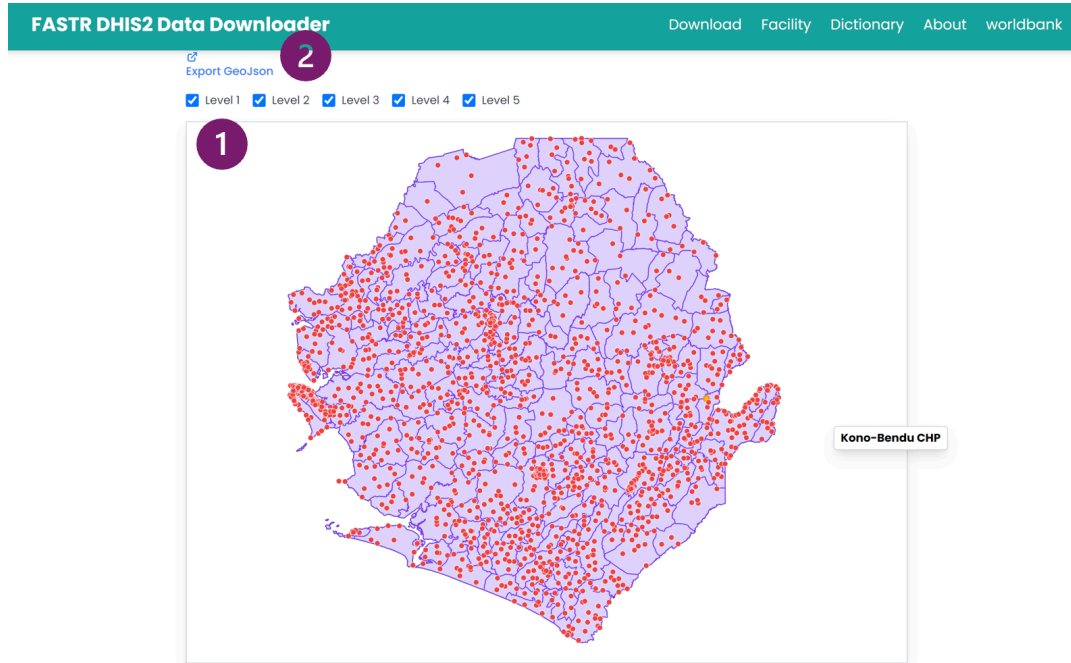
Facility management

- View complete facility list
- Filter by administrative level
- Search by facility name
- Export facility data

Data Downloader: Facility map

Geographic visualization

- Download GeoJSON boundary files
- Toggle administrative boundaries by level (Level 1 = country, Level 2 = regions, etc.)
- Higher levels display facility points
- Useful for verifying geographic structure



Data extraction

The FASTR analytics platform contains a direct import feature to automatically import data from DHIS2. This is often the easiest approach once indicators have been identified for inclusion in the platform.

 Platform import

Session 4: Getting into the platform

Introduction to the FASTR Analytics Platform

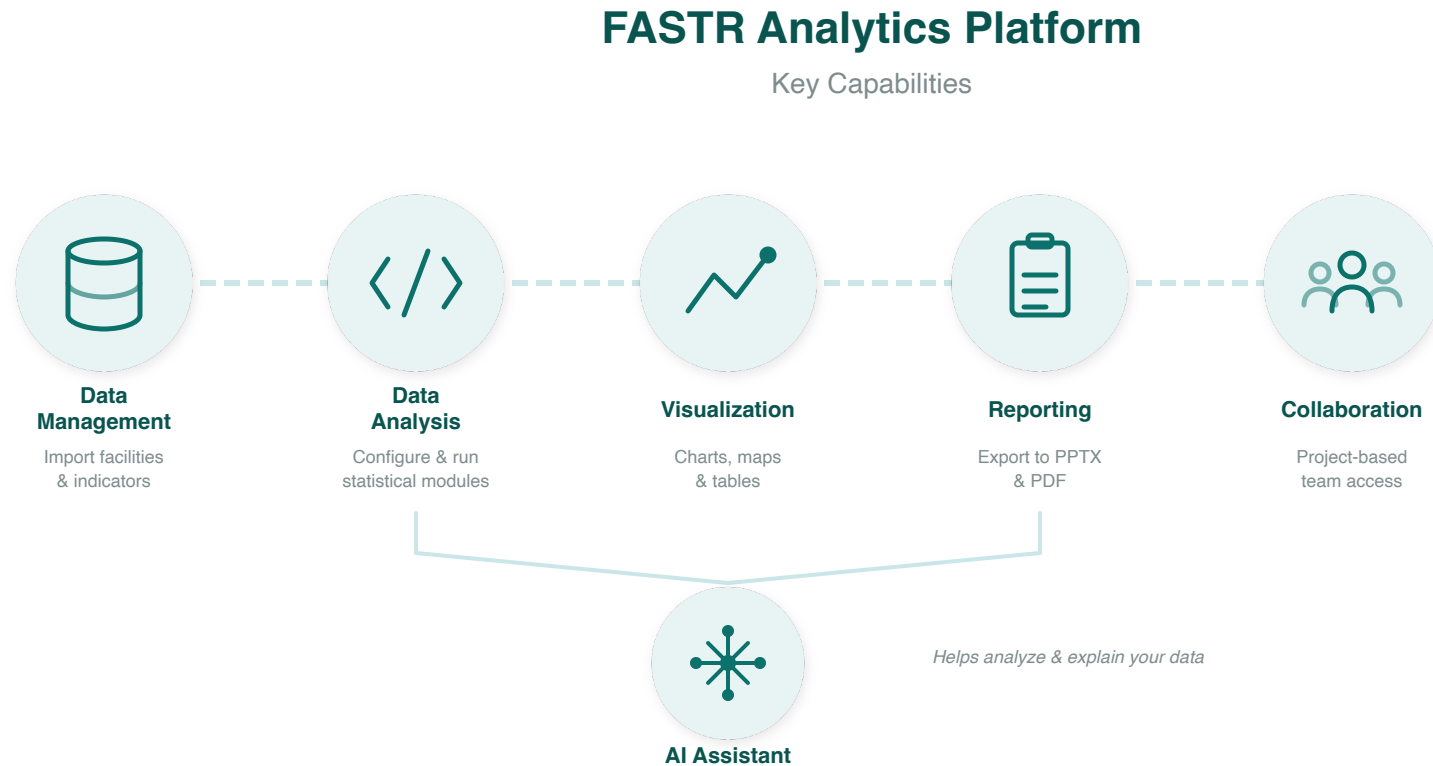
The FASTR analytics platform is a web-based tool for data quality assessment, adjustment, and analysis of routine health data.

Key features:

- Upload and analyze data from DHIS2 and other sources
- Built-in statistical methods for data quality adjustment
- User-friendly interface for running analyses
- Flexible visualization and export options

In this session, we will provide a conceptual walkthrough of the platform and its capabilities.

Platform Capabilities



Data flows from import through analysis to shareable outputs.

Country Instance

Each country has its own **instance** of the FASTR analytics platform.

An instance contains:

- All registered users and their accounts
- The shared administrative structure (regions, districts, facilities)
- Indicator definitions and data sources
- All projects created for that country

Think of an instance as your country's dedicated workspace.

User Roles and Permissions

There are two levels of permissions in the platform:

Instance-level roles:

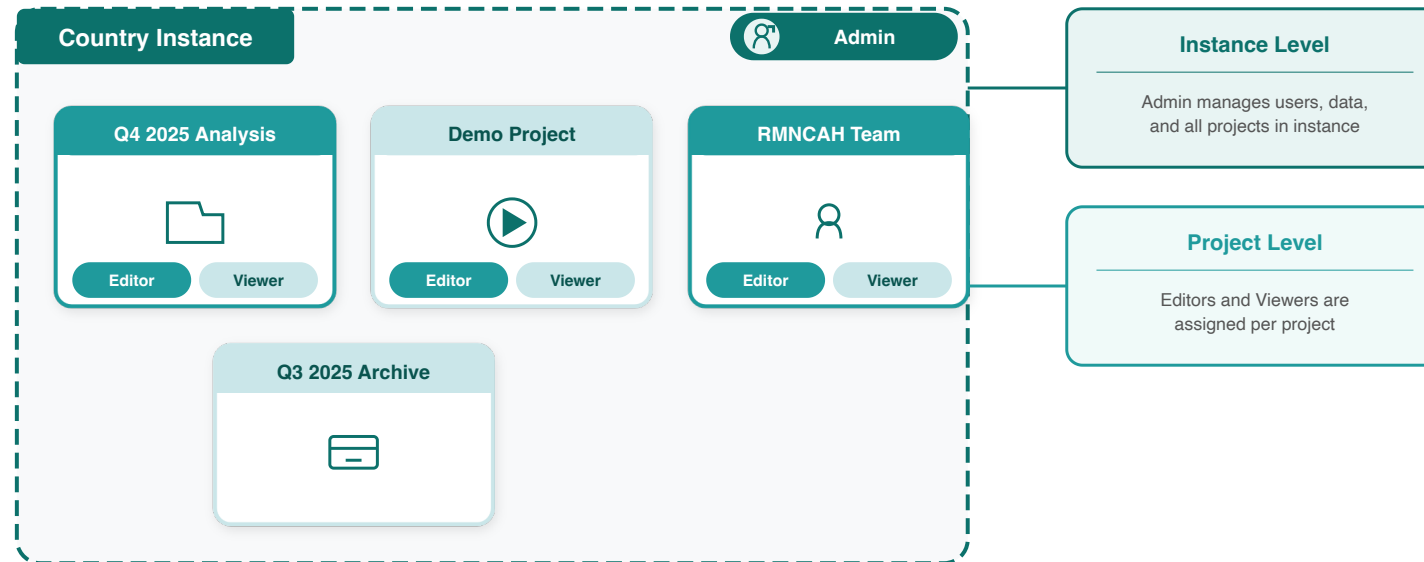
- **Instance Administrators** can add users, create projects, assign roles, upload data, import and configure modules, and run analyses

Project-level roles:

- **Project Editors** can create visualizations, create reports, and download/export results
- **Project Viewers** can view visualizations, view reports, and download/export results

Administrators are assigned per instance; Editors and Viewers are assigned per project.

Projects Within an Instance



Each country instance can contain **multiple projects**.

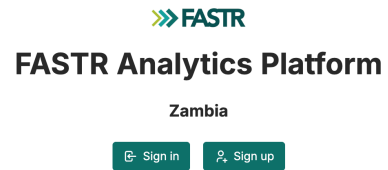
A country may only need one project, or multiple projects can be used for:

- Different versions of analyses
- A demo or playground project
- Separate projects for different teams or programs

Key questions when setting up:

- Who is the admin?
- Who can edit?
- Who can view?

Practice: Logging Into the Platform

The image shows a "Create your account" form. It includes fields for "First name", "Last name", "Email address", and "Password". A "Continue" button is at the bottom. Below the form, it says "Already have an account? Sign in".

Create your account
Welcome! Please fill in the details to get started.

First name Last name
 First name Last name

Email address
 Enter your email address

Password
 Enter your password

Continue ➤

Already have an account? [Sign in](#)

1. Go to <https://zambia.fastr-analytics.org>

3. Enter your information (verify email)

2. Click Sign up and enter your details

4. After login, you'll be added to a project

Live Demo: Platform Access & Roles



In this demo, we will:

- Navigate to the FASTR platform
- Explore user roles: Administrator, Editor, Viewer
- Review user management and permissions
- Understand the workflow for uploading data and making analytical decisions

Facilitator will demonstrate in the live platform



Lunch Break

90 minutes

Back at 14:00

Session 5: Configuring the FASTR analytics platform

Activity: Setting Up Admin Areas



In this hands-on session, we will configure:

- Admin areas (regions, districts)
- Facility structure
- Indicator definitions

Participants will work directly in the platform

Activity: Importing Data



In this hands-on session, we will:

- Review data format requirements
- Walk through the import process
- Handle validation and error checking

Participants will import their country's data

Activity: Installing and Running Modules



In this hands-on session, we will:

- Review available analysis modules
- Install required modules
- Run initial analyses

Participants will configure and run modules on their data

Day 1 Key Messages

Today we covered:

- Overview of the FASTR approach and methodology
- Data extraction methods and tools for DHIS2
- Introduction to the FASTR Analytics Platform
- Initial platform configuration for Zambia

Day 2

Recap

On Day 1, we covered:

- The FASTR approach and rapid-cycle analytics methodology
- Data extraction from DHIS2 using the Data Downloader
- Introduction to the FASTR Analytics Platform
- Platform configuration: admin areas, data import, modules

Day 2 Focus

Today we will:

- Explore FASTR methods for data quality and analysis
- Create and configure projects
- Build visualizations
- Generate reports

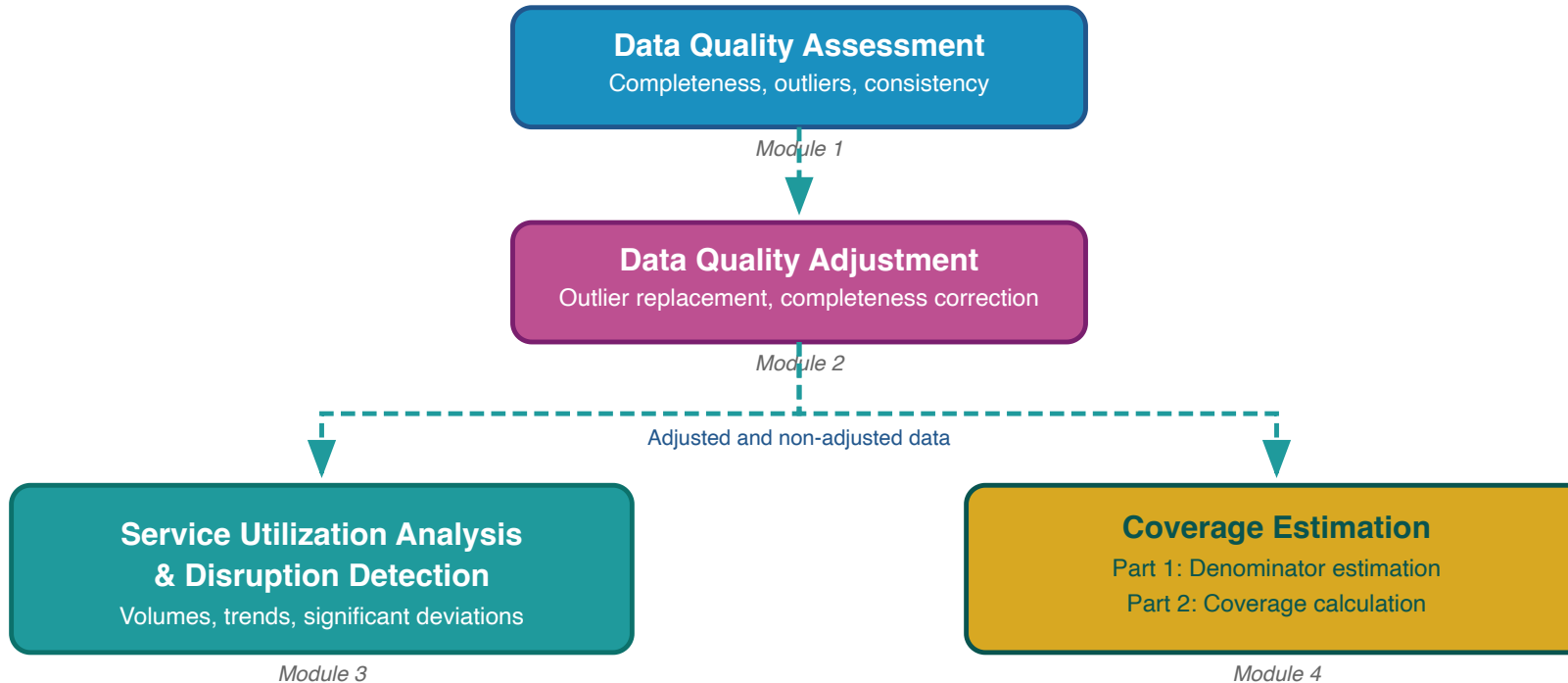
Agenda

Day 2 -- Building the Analysis: Applying FASTR Methods and Generating Outputs

Time	Agenda	Facilitator/Presenter
09:00–09:15	Recap and Day 2 agenda	GFF FASTR team
Session 6: Overview of FASTR methods and analytical outputs		
09:15–10:15	Data quality, service utilization, coverage	GFF FASTR team
Session 7: Creating a project		
10:15–11:15	Project creation and settings	GFF FASTR team
Session 8: Creating visualizations		
11:15–12:30	Creating and editing visualizations	GFF FASTR team
12:30–14:00	<i>Lunch Break</i>	
Session 9: Creating reports		
14:30–16:30	Practice creating and editing reports	GFF FASTR team

Session 6: Overview of FASTR methods and analytical outputs

FASTR analytical pipeline



The FASTR analysis follows a sequential workflow where each step builds on the previous:

- 1. Assess data quality** - Identify issues with completeness, outliers, and consistency
- 2. Adjust for quality issues** - Apply corrections to improve data reliability
- 3. Analyze adjusted data** - Generate service utilization and coverage estimates

Data quality assessment – Module 1

Evaluating the reliability of routine health information system data

Rationale for data quality assessment

Challenge: Routine health facility data may contain quality limitations:

- Reported values may fall outside plausible ranges
- Reporting gaps affect data completeness
- Inconsistencies exist between related indicators

Implications: Data quality limitations affect decision-making

- Inaccurate assessments of service delivery trends
- Misidentification of areas requiring intervention
- Suboptimal resource allocation

Objectives of data quality assessment

Objective 1: Enable analytical adjustment

Systematic data quality assessment supports the application of targeted adjustments, enhancing the utility of HMIS data for evidence-based decision-making.

Objective 2: Monitor data quality trends

Data quality assessment enables ongoing monitoring to:

- Inform indicator selection based on quality profiles across the HMIS
- Guide targeted data quality interventions and supportive supervision in areas with weaker data quality
- Evaluate the effectiveness of data quality improvement initiatives over time

Core dimensions of data quality

1. Completeness

Are health facilities submitting reports consistently?

2. Outlier prevalence

Are reported values within plausible ranges?

3. Internal consistency

Do related indicators demonstrate expected relationships?

These three dimensions provide a comprehensive assessment of data reliability for analytical purposes.

Indicator completeness

What it measures: The extent to which facilities report data on selected core indicators

Why it matters:

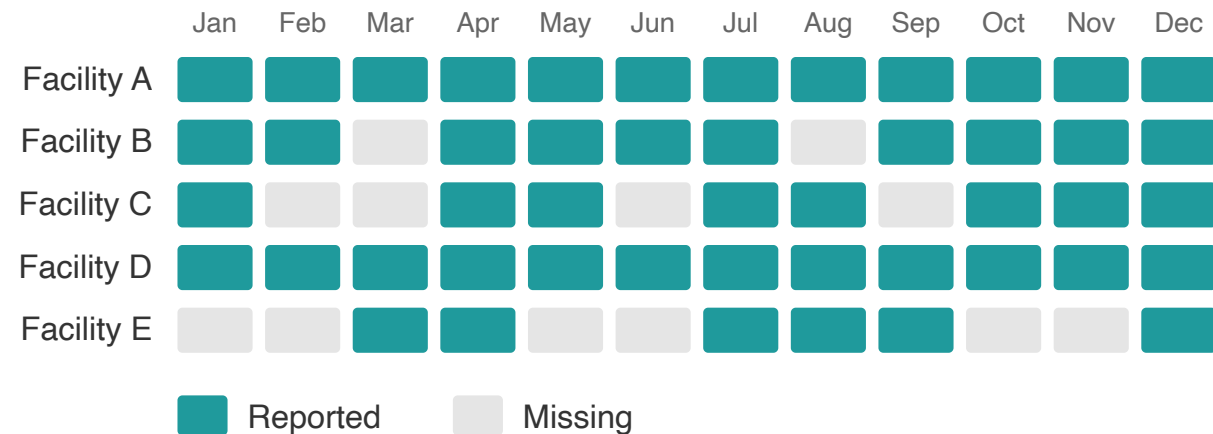
- Higher completeness improves data reliability
- Stability over time strengthens trend analysis

Key distinction:

Indicator completeness $\neq$ reporting completeness. This metric examines specific data elements, not just whether the monthly form was submitted.

Region A: Indicator Completeness

5 health facilities reporting monthly on ANC1



5 facilities $\times$ 12 months = 60 expected
48 reported $\rightarrow$ 80% completeness

Definition of indicator completeness

For the FASTR analysis, completeness is defined as:

The percentage of reporting facilities each month out of the total number of facilities expected to report.

- A facility is deemed to be "reporting" if there is a non-missing, non-zero value recorded for the indicator and month
- A facility is expected to report if it has reported any volume for that indicator anytime within a year
- Facilities that do not report for six or more consecutive months at the beginning or end of their reporting period are classified as **inactive** rather than incomplete. This prevents penalizing facilities that have not yet begun reporting or have permanently ceased operations

Notes on completeness

- A high level of completeness does not necessarily indicate that the HMIS is representative of all service delivery in the country as some services may not be delivered in facilities, or some facilities may not report
- For countries where the DHIS2 system does not store 0's, indicator completeness may be underestimated if there are many low-volume facilities for a given indicator

Completeness: Percent of monthly values that are complete

For a given indicator in a given time period, the percent of monthly values that are complete:

% complete = # monthly values that are complete / total N of monthly values

Indicator Completeness

Percentage of facility-months with complete data, Jan 2022 to Apr 2025

	Antenatal care 1	Antenatal care 4	Institutional delivery	Postnatal care 1 (newborns)	Postnatal care 1 (mothers)	BCG vaccine	Penta vaccine 1	Penta vaccine 3	Outpatient visit
District 001	93.9%	90.3%	83.3%	70.8%	82.8%	92.3%	91.9%	90.4%	94.9%
District 002	88.5%	85.9%	82.0%	64.2%	79.8%	90.8%	88.6%	88.1%	88.4%
District 003	89.8%	87.0%	85.4%	53.4%	78.5%	86.2%	86.3%	85.5%	88.3%
District 004	90.7%	82.6%	84.5%	67.6%	82.1%	89.1%	90.4%	89.7%	90.4%
District 005	89.7%	84.0%	81.3%	60.4%	80.9%	84.3%	84.2%	83.6%	89.5%
District 006	85.5%	79.4%	74.5%	56.8%	74.5%	85.3%	82.9%	79.2%	85.6%
District 007	97.2%	93.8%	91.8%	73.5%	90.8%	91.3%	95.2%	87.9%	97.0%
District 008	93.1%	87.6%	85.5%	44.6%	71.8%	79.9%	87.2%	87.5%	95.0%
District 009	95.7%	75.8%	85.0%	34.9%	77.6%	83.5%	92.2%	78.1%	95.9%
District 010	99.2%	99.6%	95.8%	98.4%	94.2%	94.6%	93.2%	88.7%	100.0%
District 011	98.0%	95.2%	94.4%	66.6%	91.7%	96.1%	95.4%	91.8%	98.2%
District 012	92.6%	86.4%	83.7%	64.5%	84.7%	84.7%	87.2%	86.7%	92.6%
District 013	93.4%	90.8%	90.7%	77.1%	88.2%	90.3%	90.8%	87.7%	93.1%
District 014	88.8%	79.1%	81.2%	77.9%	81.2%	90.0%	89.8%	86.3%	91.2%
District 015	93.0%	88.2%	85.3%	63.9%	83.1%	86.7%	88.8%	88.7%	93.1%
District 016	96.0%	85.1%	92.0%	65.7%	91.6%	96.1%	95.5%	88.6%	96.8%
District 017	90.1%	84.8%	90.4%	53.4%	83.0%	72.7%	79.3%	76.0%	89.5%
District 018	97.1%	95.9%	93.2%	79.8%	92.3%	95.4%	95.8%	92.9%	97.9%

90% or above

80% to 89%

Below 80%

Higher completeness improves the reliability of the data, especially when completeness is stable over time. Completeness is defined as the percentage of reporting facilities each month out of the total number of facilities expected to report. A facility is expected to report if it has reported any volume for each indicator anytime within a year. A high completeness does not indicate that the HMIS is representative of all service delivery in the country, as some services may not be delivered in facilities, or some facilities may not report.

Outliers

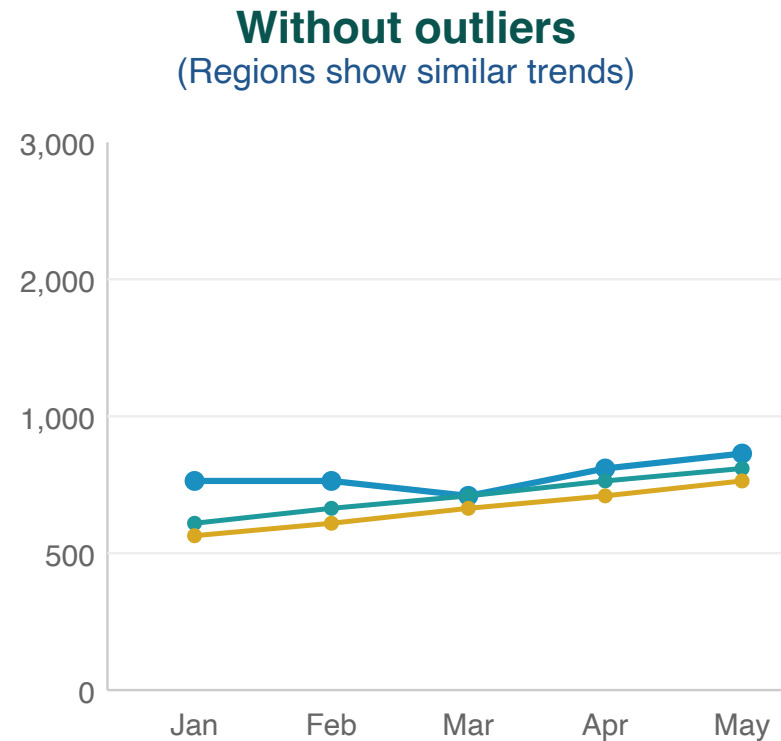
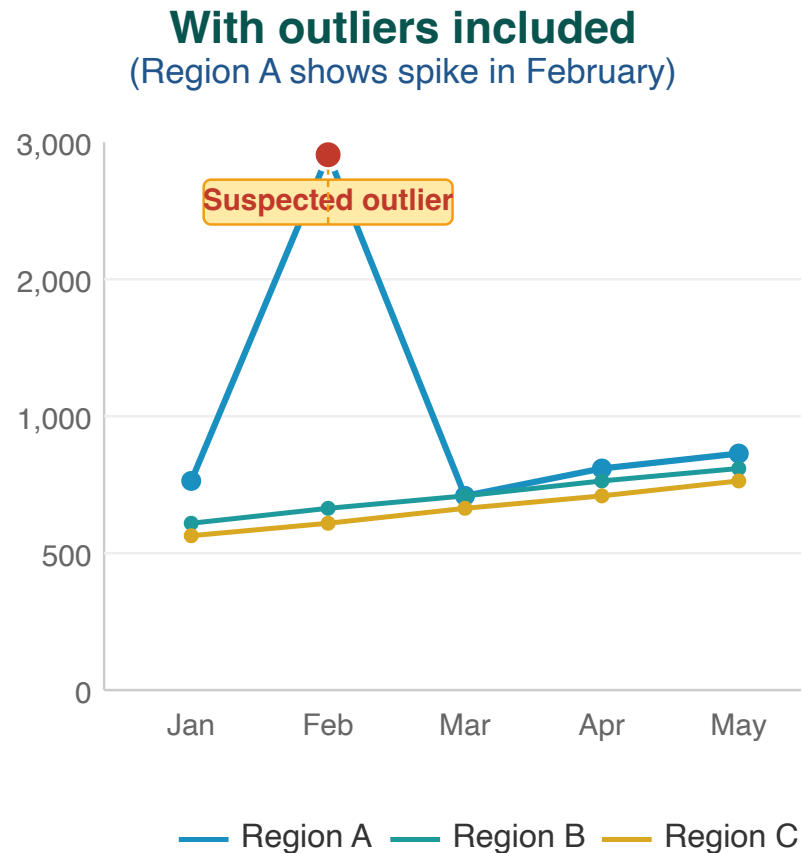
The presence of outliers examines whether a data point in a series of values is extreme (either abnormally high or low) in relation to others in the series.

Outliers can be the result of changes in programmatic activities (such as an intensified campaign) or can be data quality problems.

For the FASTR analysis, we identify outliers which are suspiciously high values compared to the usual volume of services reported by the facility (e.g., low values are not identified as outliers in the FASTR analysis).

Outlier illustration

Region A displays an anomalous spike in February that substantially exceeds values reported by other regions — indicative of a data entry error or reporting issue.



Outliers can distort the true picture of service trends

Outlier detection methodology

Outliers are identified by assessing the within-facility variation in monthly reporting for each indicator.

An outlier is defined as:

A value greater than **10 times the median absolute deviation (MAD)** from the monthly median value for the indicator in each time period, **OR** a value for which the proportional contribution in volume for a facility, indicator, and time period is **greater than 80%**

AND for which:

- The volume is **greater than or equal to the median**
- The volume is **not missing**
- The volume is **greater than 100**

Outliers: Percent of monthly values that are outliers

For a given indicator in a given time period, the percent of monthly values that are outliers:

% outliers = # monthly values that are outliers / total N of monthly values

Outliers

Percentage of facility-months that are outliers, May 2024 to Apr 2025

	Antenatal care 1	Antenatal care 4	Institutional delivery	Postnatal care 1 (newborns)	Postnatal care 1 (mothers)	BCG vaccine	Penta vaccine 1	Penta vaccine 3	Outpatient visit
Region 001	0.6%	0.2%	0.2%	1.9%	1.2%	0.6%	0.9%	0.6%	0.9%
Region 002	0.6%	0.0%	1.2%	1.0%	1.8%	0.2%	0.2%	0.1%	2.7%
Region 003	0.5%	0.4%	0.1%	0.0%	0.1%	0.1%	0.4%	0.6%	1.4%
Region 004	0.4%	0.0%	0.0%	0.8%	0.0%	0.5%	0.8%	0.9%	0.1%
Region 005	0.5%	0.9%	0.5%	0.8%	0.5%	0.6%	0.8%	0.3%	3.3%
Region 006	0.4%	0.0%	0.1%	0.3%	0.1%	1.3%	2.3%	2.5%	0.6%

- 3% or above
- 1% to 2%
- Below 1%

Outliers are reports which are suspiciously high compared to the usual volume reported by the facility in other months. Outliers are identified by assessing the within-facility variation in monthly reporting for each indicator. Outliers are defined observations which are greater than 10 times the median absolute deviation (MAD) from the monthly median value for the indicator in each time period, OR a value for which the proportional contribution in volume for a facility, indicator, and time period is greater than 80%. Outliers are only identified for indicators where the volume is greater than or equal to the median, the volume is not missing, and the average volume is greater than 100.

Consistency between related indicators

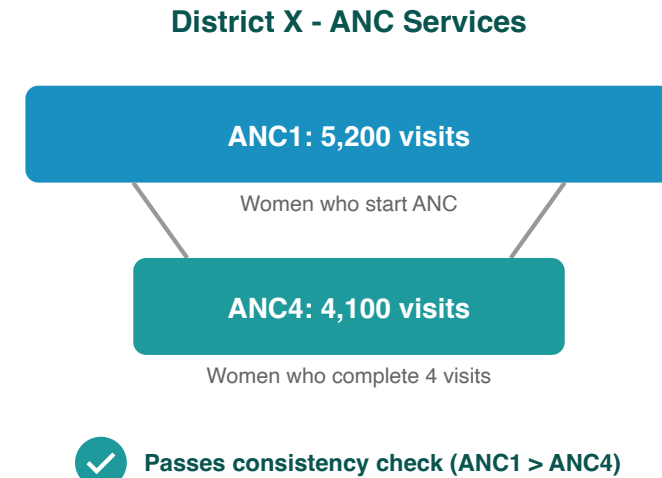
Program indicators with a predictable relationship are examined to determine whether the expected relationship exists between them. In other words, this process examines whether the observed relationship between the indicators, as shown in the reported data, is that which is expected.

Indicator pairs assessed

Indicator pair	Expected relationship
ANC1 / ANC4	Ratio should be ≥ 0.95
Penta1 / Penta3	Ratio should be ≥ 0.95
BCG / Facility delivery	Within 30% (≥ 0.7 and ≤ 1.3)

These pairs have expected relationships. We expect $ANC1 > ANC4$ since not all women complete four visits.

BCG is a birth dose vaccine so we expect similar numbers to facility deliveries, with a 30% tolerance for variability.

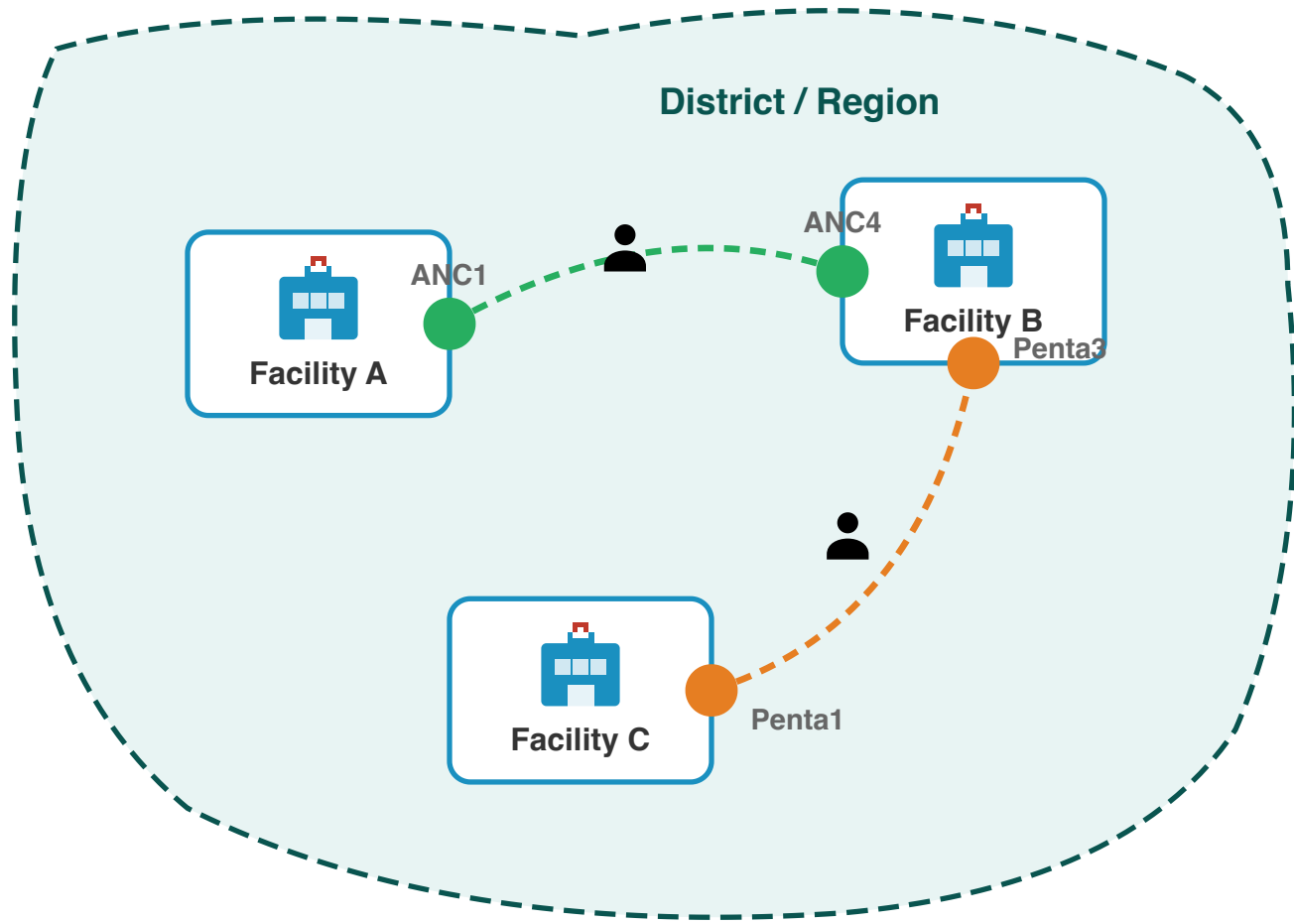


Why assess consistency at district level?

Patients often access different services at different facilities within a district:

- A woman may attend **ANC1** at a nearby health post, but travel to a health centre for **ANC4**
- A child may receive **Penta1** at a local clinic, but complete **Penta3** at a district hospital

Checking consistency at the facility level would miss these patterns. Aggregating to district level captures the complete picture of service utilization within a geographic area.



Internal consistency: FASTR output

Internal consistency

Percentage of sub-national areas meeting consistency benchmarks, May 2024 to Apr 2025

	ANC1 is larger than ANC4	Delivery is approximately equal to BCG	Penta1 is larger than Penta3
Region 001	100.0%	0.0%	97.2%
Region 002	100.0%	0.0%	100.0%
Region 003	100.0%	45.8%	83.3%
Region 004	100.0%	37.5%	87.5%
Region 005	100.0%	0.0%	87.5%
Region 006	100.0%	49.0%	84.4%

- 90% or above
- 80% to 89%
- Below 80%

Internal consistency assesses the plausibility of reported data based on related indicators. Consistency metrics are approximate - depending on timing and seasonality, indicator definitions, and the nature of service delivery and reporting, values may be expected to sit outside plausible ranges. Indicators which are similar are expected to have roughly the same volume over the year (within a 30% margin). The data in this analysis is adjusted for outliers.

Data quality summary score

A composite measure of data quality provides an overall view of how well a dataset meets quality standards.

By integrating multiple dimensions of data quality into a single score, it simplifies the interpretation of detailed information from several measures. This allows health systems to quickly assess the reliability of data, making it easier to identify trends and issues at a glance.

Definition of adequate data quality

For the FASTR analysis, we defined adequate data quality as:

- No missing indicator data for OPD, Penta1, and ANC1, where available, **AND**
- No outliers for OPD, Penta1, and ANC1, where available, **AND**
- Consistent reporting between Penta1/Penta3 and ANC1/ANC4

Overall DQA score: Percent of monthly values meeting all criteria

For a given indicator in a given time period, the percent of monthly values meeting all DQA criteria:

% adequate quality = # monthly values meeting all criteria / total N of monthly values

Overall DQA score

Percentage of facility - months with adequate data quality over time

	2022	2023	2024	2025
Region 001	60.8%	76.6%	76.4%	84.4%
Region 002	55.3%	61.2%	58.2%	52.0%
Region 003	69.3%	73.4%	60.6%	47.0%
Region 004	54.6%	70.7%	70.0%	84.9%
Region 005	57.9%	69.5%	56.6%	55.4%
Region 006	74.0%	84.7%	72.2%	71.7%

80% or above

70% to 79%

Below 70%

Adequate data quality is defined as: 1) No missing data or outliers for OPD, Penta1, and ANC1, where available 2) Consistent reporting between Penta1/Penta3 and ANC1/ANC4.

Mean DQA score: How close are we to adequate quality?

The mean DQA score shows how close a facility's data is to meeting all quality criteria. A score of **100%** means the data passes all DQA checks—no missing values, no outliers, and consistent reporting.

Mean DQA = (completeness & outlier score + consistency score) / 2

Mean DQA score
Average data quality score across facility - months

	2022	2023	2024	2025
Region 001	90.2%	95.5%	95.5%	96.5%
Region 002	87.3%	89.0%	88.2%	85.6%
Region 003	92.8%	94.8%	91.0%	84.3%
Region 004	88.0%	93.9%	93.0%	97.0%
Region 005	89.5%	93.6%	91.1%	88.6%
Region 006	93.2%	96.3%	93.4%	93.4%

- 80% or above
- 70% to 79%
- Below 70%

Items included in the DQA score include: No missing data for 1) OPD, 2) Penta1, and 3) ANC1, where available; No outliers for 4) OPD, 5) Penta1, and 6) ANC1, where available; Consistent reporting between 7) Penta1/Penta3, 8) ANC1/ANC4, 9)BCG/Delivery, where available.

DQA module: Configuration parameters

Parameter	Description
Proportion threshold for outlier detection	Adjusts the threshold for proportional contribution to flag a facility-month as an outlier
Minimum count threshold for consideration	Defines the minimum count required for a facility-month to be considered an outlier
Number of MADs	Outliers are defined as observations which are greater than X times the median absolute deviation (MAD) from the monthly median value for the indicator in each time period
Indicators subjected to DQA	Defines which indicators are included for assessment of outliers and completeness for inclusion in the DQA score
Consistency pairs used	Defines which indicator pairs are used for consistency analysis and the expected ratio ranges

Data quality adjustment – Module 2

Correcting outliers and imputing missing values to improve data reliability

Rationale for data quality adjustment

Routine HMIS data contain two common limitations that can distort analytical results:

Issue	Impact on analysis
Outliers	Extreme values create artificial spikes in service volumes
Incomplete reporting	Missing data creates artificial declines that do not reflect actual service delivery

FASTR addresses these limitations by replacing problematic values with estimates derived from each facility's historical reporting patterns.

Adjustment scenarios

To support transparency and sensitivity analysis, FASTR produces four parallel datasets:

Scenario	Description
Unadjusted	Original reported values
Outliers adjusted	Extreme values replaced
Completeness adjusted	Missing values imputed
Both adjusted	All corrections applied

Indicators excluded from adjustment

Certain indicators are excluded from the adjustment process:

- **Mortality indicators** (maternal deaths, neonatal deaths, under-5 deaths): These represent discrete events where smoothing or imputation is not appropriate
- **Low-volume indicators**: Indicators that never exceed 100 reported events in any month are excluded from adjustment

Outlier adjustment methodology

Outlier values are replaced using facility-specific historical data. The adjustment follows a hierarchical approach:

Priority	Method	Application
1	Centered 6-month average	3 months before + 3 months after the outlier
2	Forward 6-month average	When insufficient preceding data (e.g., start of series)
3	Backward 6-month average	When insufficient following data (e.g., end of series)
4	Same month, previous year	When rolling averages unavailable; useful for seasonal indicators
5	Facility historical mean	Mean of all valid values for this indicator at this facility

Outlier adjustment: FASTR output

Deviance Due to Outliers

Percent change in volume due to outlier adjustment, Oct 2024 to Sep 2025

	Antenatal client 1st visit	Antenatal client 4th visit	Institutional delivery (normal, assisted, and csection)	Postnatal care 1 (newborns)	Postnatal care 1 (mothers)	BCG dose	Pentavalent 1st dose	Pentavalent 3rd dose	Family planning methods- long acting	Family planning methods- short acting	Fever case tested positive for malaria (RDT and microscopy)	Malaria treated with ACT
Bo District	0.0%	0.2%	4.6%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.2%	0.2%
Bombali District	1.3%	0.0%	0.0%	2.3%	0.0%	0.0%	0.0%	0.0%	0.0%	0.8%	0.0%	0.0%
Bonthe District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	2.5%	0.0%	0.0%	0.0%
Falaba District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.7%	0.0%	0.0%	2.2%	0.0%	0.0%
Kailahun District	0.0%	0.0%	0.6%	0.0%	0.0%	0.0%	0.0%	0.0%	4.5%	2.1%	0.2%	0.2%
Kambia District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.8%	0.0%	0.0%	0.0%
Karene District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	2.7%	1.3%	5.7%	4.7%	0.0%	0.0%
Kenema District	0.0%	0.0%	0.0%	0.0%	0.0%	1.2%	1.7%	1.4%	1.8%	0.4%	0.0%	0.0%
Koinadugu District	0.0%	0.0%	0.0%	0.0%	0.3%	0.0%	0.0%	0.0%	2.6%	1.9%	0.0%	0.0%
Kono District	0.7%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	3.2%	0.6%	2.0%	1.8%
Moyamba District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	3.7%	0.4%	0.1%	0.1%
Port Loko District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.8%	0.0%	0.0%	0.0%
Pujehun District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.3%	0.3%
Tonkolili District	0.0%	0.3%	0.0%	0.0%	0.0%	0.6%	0.0%	0.0%	1.5%	1.4%	0.1%	0.1%
Western Area Rural District	0.0%	0.0%	0.7%	1.5%	1.4%	0.5%	1.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Western Area Urban District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.6%	0.9%	1.1%	0.5%	1.5%	0.3%

3% or above

1% to 2%

Below 1%

Outliers are reports which are suspiciously high compared to the usual volume reported by the facility in other months. Outliers are identified by assessing the within-facility variation in monthly reporting for each indicator. Outliers are defined observations which are greater than 10 times the median absolute deviation (MAD) from the monthly median value for the indicator in each time period, OR a value for which the proportional contribution in volume for a facility, indicator, and time period is greater than 80%. Outliers are only identified for indicators where the volume is greater than or equal to the median, the volume is not missing, and the average volume is greater than 100. The deviance is the difference in volume after removing the outlier. High levels of deviance can affect the plausibility of the data.

Completeness adjustment methodology

For months identified as incomplete or missing, values are imputed using the same 6-month rolling average approach applied to outlier adjustment.

Priority	Method	Application
1	Centered 6-month average	When sufficient data exists before and after the gap
2	Forward 6-month average	For gaps at the start of the time series
3	Backward 6-month average	For gaps at the end of the time series
4	Facility historical mean	Mean of all valid values for this indicator at this facility

This approach prevents temporary reporting gaps from creating artificial declines in service volumes.

Completeness adjustment: FASTR output

Deviance Due to Incompleteness

Percent change in volume due to completeness adjustment, Oct 2024 to Sep 2025

	Antenatal client 1st visit	Antenatal client 4th visit	Institutional delivery (normal, assisted, and csection)	Postnatal care 1 (newborns)	Postnatal care 1 (mothers)	BCG dose	Pentavalent 1st dose	Pentavalent 3rd dose	Family planning methods- long acting	Family planning methods- short acting	Fever case tested positive for malaria (RDT and microscopy)	Malaria treated with ACT
Bo District	2.2%	4.7%	3.1%	4.7%	3.8%	2.2%	7.2%	9.0%	12.3%	18.4%	3.6%	8.3%
Bombali District	2.6%	3.3%	4.1%	10.4%	12.9%	7.4%	5.9%	5.6%	13.0%	5.6%	10.8%	8.6%
Bonthe District	1.5%	5.2%	3.6%	8.3%	4.3%	1.6%	4.1%	4.0%	24.3%	8.3%	2.7%	6.1%
Falaba District	0.5%	2.8%	3.0%	6.4%	4.2%	0.9%	0.3%	0.3%	12.8%	7.5%	1.8%	4.7%
Kailahun District	0.5%	1.9%	1.3%	13.8%	11.9%	0.5%	0.4%	0.4%	8.3%	8.1%	2.8%	4.4%
Kambia District	0.8%	2.8%	0.4%	1.4%	1.3%	0.5%	0.6%	0.8%	9.6%	5.3%	0.3%	1.4%
Karene District	0.3%	0.8%	0.0%	0.0%	0.0%	0.0%	0.0%	0.3%	8.3%	6.3%	2.3%	4.8%
Kenema District	0.3%	1.2%	0.7%	1.7%	1.3%	0.1%	0.2%	0.2%	5.6%	4.5%	0.5%	1.1%
Koinadugu District	2.3%	3.9%	2.7%	4.3%	2.6%	1.5%	2.1%	2.3%	41.1%	23.9%	13.1%	14.9%
Kono District	0.4%	2.1%	1.2%	5.6%	3.1%	0.6%	0.7%	0.9%	6.8%	7.3%	1.2%	6.5%
Moyamba District	1.4%	3.8%	2.9%	4.0%	3.3%	1.7%	2.0%	2.2%	13.9%	7.4%	2.8%	5.3%
Port Loko District	1.5%	2.0%	2.1%	2.9%	4.8%	1.6%	1.3%	1.4%	19.9%	9.8%	10.3%	8.8%
Pujehun District	0.3%	1.1%	2.1%	5.2%	4.0%	0.1%	0.2%	0.3%	7.0%	3.9%	0.3%	0.9%
Tonkolili District	1.2%	3.0%	4.6%	6.5%	2.7%	1.5%	1.8%	1.7%	9.6%	8.2%	2.5%	3.7%
Western Area Rural District	0.7%	0.8%	1.7%	1.7%	2.0%	0.6%	2.6%	2.4%	22.7%	10.1%	0.2%	0.6%
Western Area Urban District	6.2%	8.6%	5.7%	15.9%	13.5%	5.7%	6.6%	6.4%	19.9%	15.5%	4.5%	6.4%

3% or above

1% to 2%

Below 1%

Completeness is defined as the percentage of reporting facilities each month out of the total number of facilities expected to report. A facility is expected to report if it has reported any volume for each indicator anytime within a year. The deviance is the difference in volume after imputing incomplete data. High levels of deviance can affect the plausibility of the data.

Combined adjustment: FASTR output

Deviance Due to Incompleteness and Outliers

Percent change in volume due to both outlier and completeness adjustment, Oct 2024 to Sep 2025

	Antenatal client 1st visit	Antenatal client 4th visit	Institutional delivery (normal, assisted, and csection)	Postnatal care 1 (newborns)	Postnatal care 1 (mothers)	BCG dose	Pentavalent 1st dose	Pentavalent 3rd dose	Family planning methods- long acting	Family planning methods- short acting	Fever case tested positive for malaria (RDT and microscopy)	Malaria treated with ACT
Bo District	2.2%	4.5%	1.5%	4.7%	3.8%	2.2%	7.2%	9.0%	12.3%	18.4%	3.4%	8.1%
Bombali District	1.3%	3.3%	4.1%	8.1%	12.9%	7.4%	5.9%	5.6%	13.0%	4.7%	10.8%	8.6%
Bonthe District	1.5%	5.2%	3.6%	8.3%	4.3%	1.6%	4.1%	4.0%	21.8%	8.3%	2.7%	6.1%
Falaba District	0.5%	2.8%	3.0%	6.4%	4.2%	0.9%	0.4%	0.3%	12.8%	5.3%	1.8%	4.7%
Kailahun District	0.5%	1.9%	0.7%	13.8%	11.9%	0.5%	0.4%	0.4%	3.8%	5.9%	2.6%	4.2%
Kambia District	0.8%	2.8%	0.4%	1.4%	1.3%	0.5%	0.6%	0.8%	8.8%	5.3%	0.3%	1.4%
Karene District	0.3%	0.8%	0.0%	0.0%	0.0%	0.0%	2.7%	1.0%	2.6%	1.6%	2.3%	4.8%
Kenema District	0.3%	1.2%	0.7%	1.7%	1.3%	1.1%	1.5%	1.2%	3.7%	4.1%	0.5%	1.1%
Koinadugu District	2.3%	3.9%	2.7%	4.3%	2.3%	1.5%	2.1%	2.3%	38.6%	22.1%	13.1%	14.9%
Kono District	0.3%	2.1%	1.2%	5.6%	3.1%	0.6%	0.7%	0.9%	3.6%	6.8%	0.9%	4.7%
Moyamba District	1.4%	3.8%	2.9%	4.0%	3.3%	1.7%	2.0%	2.2%	10.2%	7.0%	2.7%	5.2%
Port Loko District	1.5%	2.0%	2.1%	2.9%	4.8%	1.6%	1.3%	1.4%	19.1%	9.8%	10.3%	8.8%
Pujehun District	0.3%	1.1%	2.1%	5.2%	4.0%	0.1%	0.2%	0.3%	7.0%	3.9%	0.0%	0.6%
Tonkolili District	1.2%	2.7%	4.6%	6.5%	2.7%	0.9%	1.8%	1.7%	8.1%	6.7%	2.4%	3.6%
Western Area Rural District	0.7%	0.8%	0.9%	0.2%	0.6%	0.1%	1.6%	2.4%	22.7%	10.1%	0.2%	0.6%
Western Area Urban District	6.2%	8.6%	5.7%	15.9%	13.5%	5.7%	6.0%	5.5%	18.7%	15.0%	3.0%	6.1%

3% or above
1% to 2%
Below 1%

TBD

Service utilization analysis – Module 3

Detecting and quantifying changes in health service delivery over time

Objectives

1. Measure changes in service volume

Year-over-year comparison of service volumes identifies increases or decreases across regions and indicators.

2. Detect and quantify disruptions

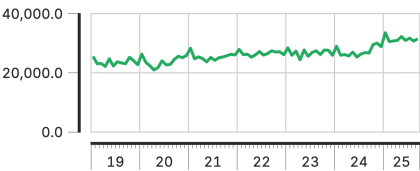
Statistical comparison of observed volumes against expected levels—derived from historical trends and seasonal patterns—enables identification and quantification of service shortfalls or surpluses.

Service utilization over time

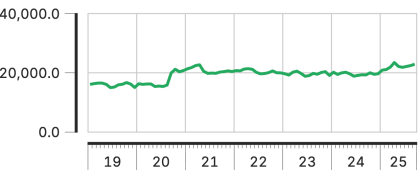
Service utilization over time

Jan 2019 to Sep 2025

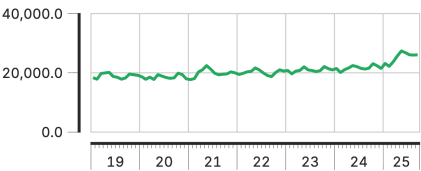
Antenatal client 1st visit



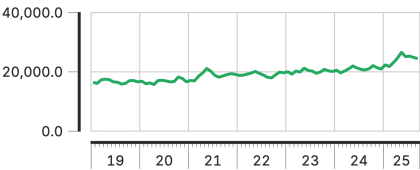
Antenatal client 4th visit



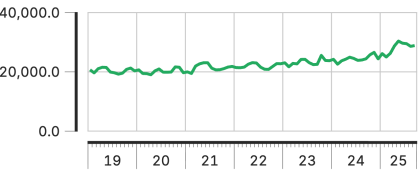
Institutional delivery (normal, assisted, and csection)



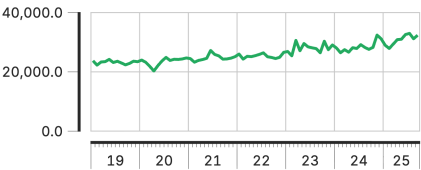
Postnatal care 1 (mothers)



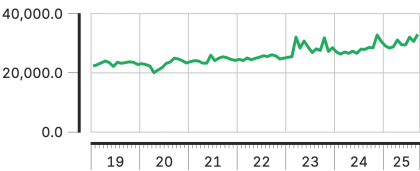
BCG dose



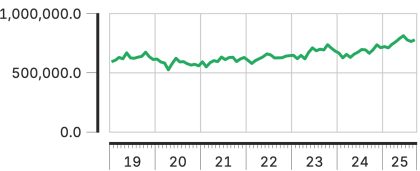
Pentavalent 1st dose



Pentavalent 3rd dose



Outpatient visit



Number of services after both outlier and completeness adjustment

Service volume is adjusted for outliers and completeness.

Year-over-year change

Year-over-year percent change quantifies shifts in service delivery between consecutive years.

For each indicator and region, total volume in the current year is compared to the previous year:

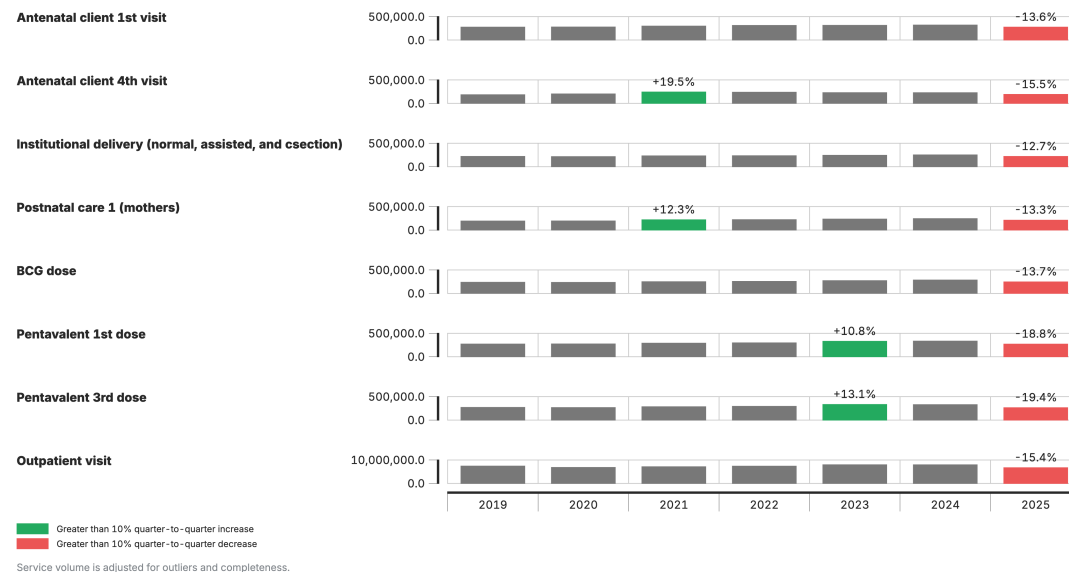
Percent change = $(\text{Current year} - \text{Previous year}) \div \text{Previous year} \times 100$

Changes exceeding **±10%** are flagged for review.

Output: Change in service volume

Service volume by year & year-on-year change

Jan 2019 to Sep 2025



Interpretation

Bars show annual service volumes by region.
Percentages indicate year-over-year change.

Key considerations:

- Which regions exhibit the largest changes?
- Are changes consistent across regions or geographically concentrated?
- Do patterns vary by indicator?

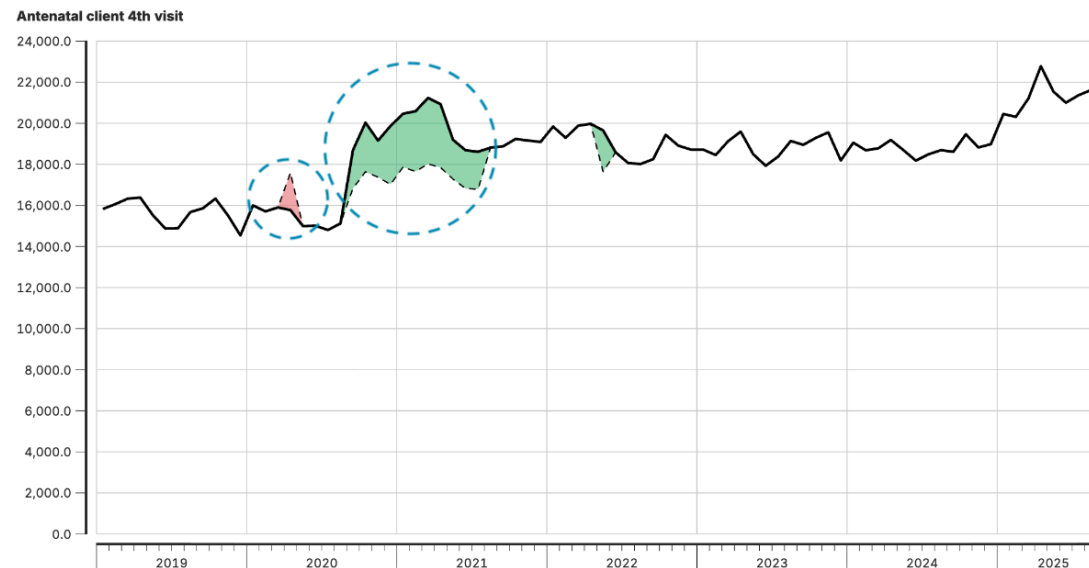
Disruption detection

Our approach to service disruptions and surpluses utilizes an interrupted time series regression with facility-level fixed effects. Previous large and unexpected changes in historical data are removed. Unexpected volume changes are estimated by comparing observed volume to expected volume based on historical trends and seasonality.

Disruptions and surpluses

Disruptions are flagged when volumes fall below anticipated levels, signaling potential barriers to access, resource shortages, or system failures.

Surpluses occur when volumes exceed expectations, which may indicate increased demand, over-reporting, or changes in service delivery.



How it works

Using past data to set expectations: We look at the past few years of data to understand the typical pattern for each month, accounting for regular seasonal changes.

Spotting unusual changes: We compare current service volumes to expectations. If we see volumes much higher or lower than expected, we flag it as an unusual change.

Handling past disruptions: We adjust historical data by removing previous big, unexpected changes so one-off events don't skew our understanding of what's "normal."

Detecting disruptions over time: We look at trends to see if there are clear shifts in health service use over several months.

Comparison to DHIS2

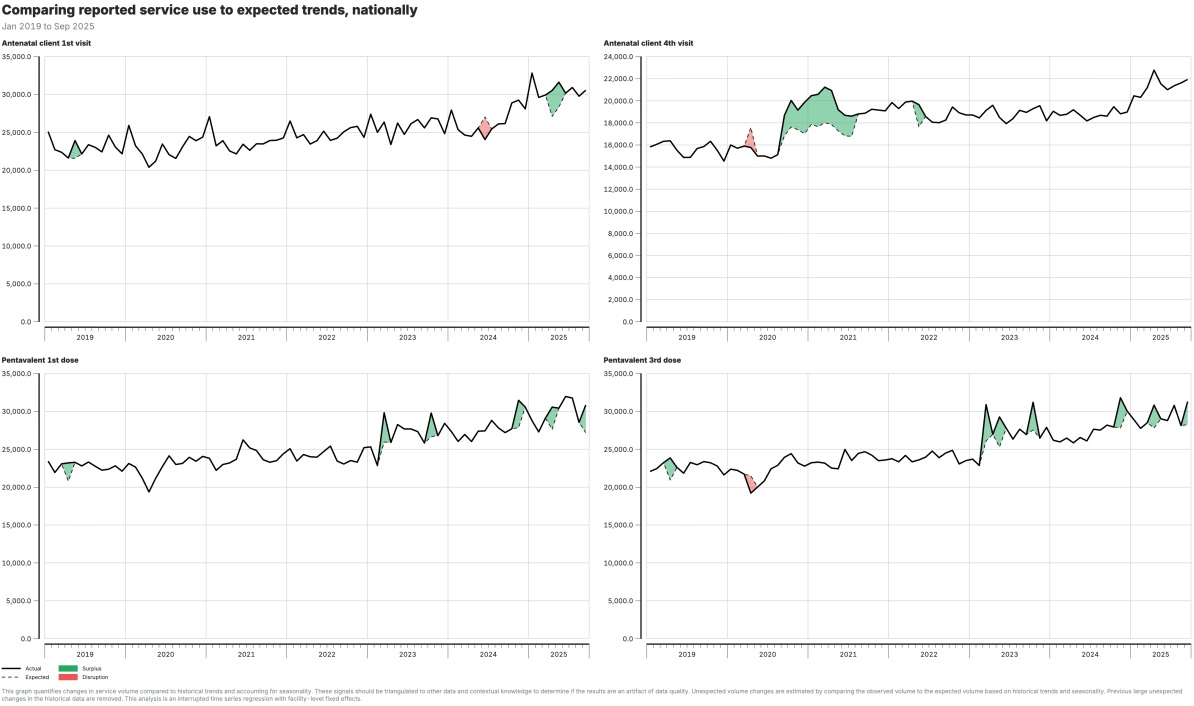
Extension of service utilization analysis, using more complex statistical approaches not available in DHIS2.

Using a regression framework, we are able to:

- Account for seasonality
- Exclude unusual changes to ensure one-off events aren't influencing normal trends
- Use historical data as a baseline for context
- Detect disruptions and recovery patterns
- Quantify changes with a robust methodology as compared to just observing simple fluctuations in a trend line

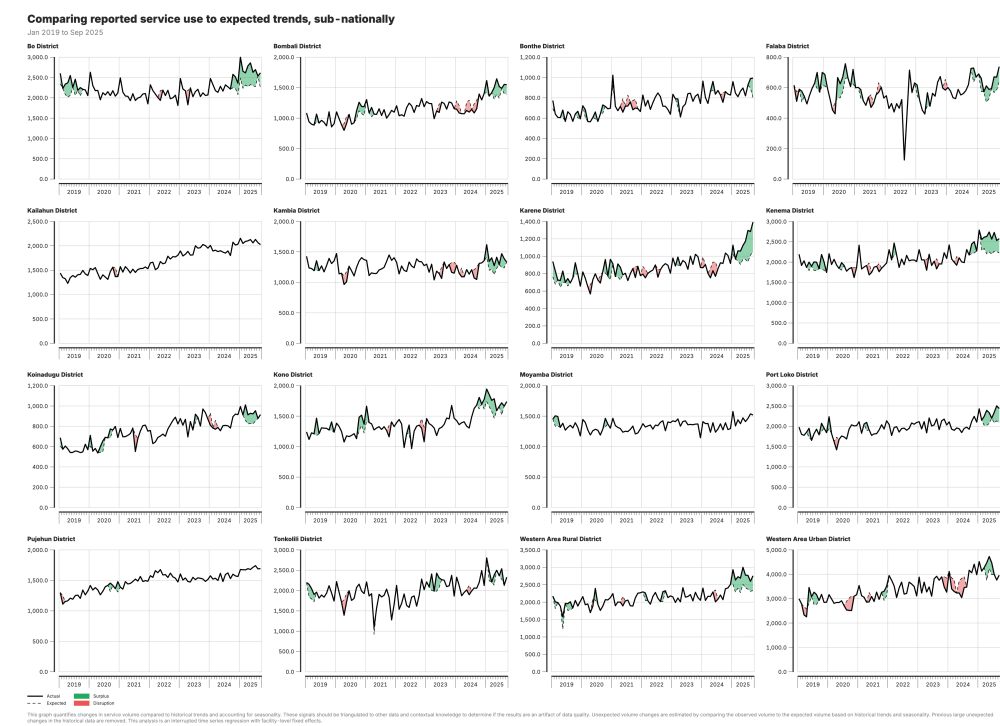
This improves the ability to interpret and compare utilization data across national and sub-national areas without needing population denominators.

Output: Actual vs expected (national)



National-level comparison of observed service volumes against expected values derived from historical trends and seasonal patterns.

Output: Actual vs expected (subnational)



Subnational disaggregation enables identification of geographic areas where disruptions are concentrated.

Service utilization module: Configuration parameters

Parameter	Description
Count variable to use for modeling	Which adjusted count to use for calculating expected values
Count variable to use for visualization	Which adjusted count to use as actual observed values
Run district-level model	Run regressions at admin_area_3 level. Set to Yes for detailed analysis, No for faster runtime
Run admin_area_4 analysis	Run finest-level analysis. Warning: can be very slow for large datasets
Threshold for MAD-based control limits	Number of MADs to flag sharp deviations. Default 1.5; higher = less sensitive
Smoothing window (k)	Window size in months for rolling median smoothing. Must be odd. Default 7
Dip threshold	Flag if actual falls below this proportion of expected. Default 0.9 ($\geq 10\%$ drop); use 0.8 to flag only big drops
Difference percent threshold	Highlight points where actual differs from expected by more than this percent. Default 10

Service coverage estimates – Module 4

Estimating the percentage of the target population that received a given health service

Our approach to service coverage analysis

Our approach derives and validates population denominators, significantly improving coverage estimates reported from HMIS systems.

In countries with accurate data, this approach helps identify subnational inequities and updates outdated estimates, while in countries with less precise data, the trends still provide valuable insights into performance.

We use these estimates to track recent trends and subnational disparities in the coverage of selected health services.

Two-part analytical process

The coverage estimation module operates in two sequential parts:

Part	Components
Part 1: Denominator calculation	Calculate target populations using multiple methods; compare against survey benchmarks; select optimal denominator for each indicator
Part 2: Coverage estimation	Apply denominator selections; project survey estimates forward using HMIS trends; generate final coverage estimates

What is service coverage?

Service coverage represents the proportion of the target population that received a specified health service.

$$\text{Service coverage} = \frac{\text{Population who received the service}}{\text{Population who need the service (target population)}}$$

Numerator comes from DHIS2 data, count of services adjusted for outliers

Several options for estimating the denominator

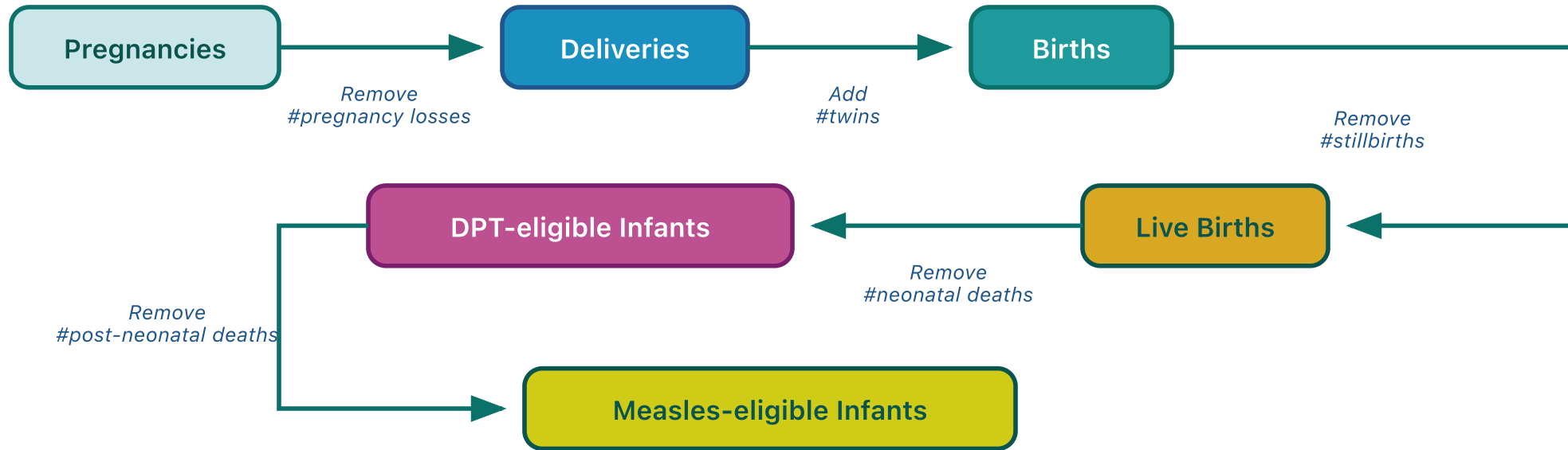
Denominators by service type

Each health indicator corresponds to a specific target population:

Service	Target population (denominator)
ANC1, ANC4	Pregnant women
Institutional delivery	Live births
BCG	Live births
Penta1, Penta3	Infants surviving beyond neonatal period
Measles1, Measles2	Infants surviving beyond infancy

Demographic cascade

Sequential demographic adjustments transform one target population estimate into another. Starting from pregnancies, demographic factors are applied to derive subsequent denominators:



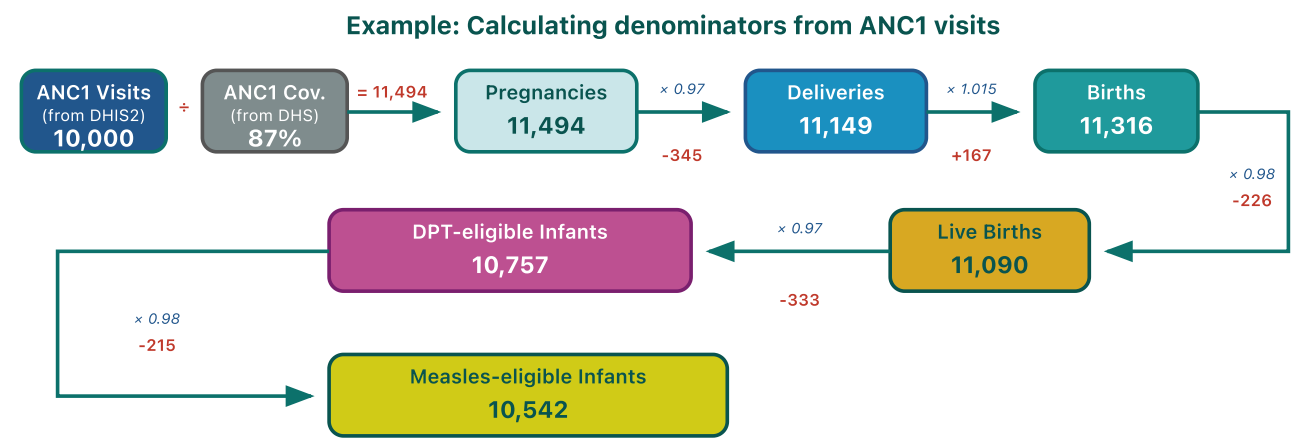
Default values:

Pregnancy loss rate = 0.03 | Twin rate = 0.015 | Stillbirth rate = 0.02 | NMR = 0.03 | PNMR = 0.02

Use country-specific values for stillbirth rate, NMR, PNMR, and IMR when available from DHS/MICS.

Denominator cascade: Illustration

Starting from ANC1 service counts, demographic adjustment factors are applied sequentially to derive denominators for other services:



Calculation summary:

Step 1: Pregnancies = ANC1 visits ÷ ANC1 coverage = 10,000 ÷ 0.87 = 11,494

Step 2: Apply demographic factors through the cascade:

Pregnancies (11,494) → Deliveries (11,149) → Births (11,316) → Live Births (11,090) → DPT-eligible (10,757) → Measles-eligible (10,542)

Rates: Pregnancy loss = 3%, Twin rate = 1.5%, Stillbirth = 2%, NMR = 3%, PNMR = 2%

Forward and backward derivation

From any entry point, the cascade derives denominators in both directions:

Direction	Method	Example from Penta1
Forward	Apply mortality/attrition rates	DPT-eligible → Measles1-eligible → Measles2-eligible
Backward	Reverse mortality rates (add deaths back)	DPT-eligible → Live births → Births → Deliveries → Pregnancies

Backward derivation enables estimation of upstream populations from downstream service counts.

Denominators by entry point

Each HMIS indicator serves as an entry point. The module derives all target populations via forward and backward cascades:

Entry point	Base calculation	Forward derivation	Backward derivation
ANC1	ANC1 ÷ coverage → Pregnancies	Deliveries → Live births → DPT-eligible → Measles-eligible	—
Deliveries	Deliveries ÷ coverage → Deliveries	Live births → DPT-eligible → Measles-eligible	Pregnancies
BCG	BCG ÷ coverage → Live births	DPT-eligible → Measles-eligible	Deliveries → Pregnancies
Penta1	Penta1 ÷ coverage → DPT-eligible	Measles1-eligible → Measles2-eligible	Live births → Births → Deliveries → Pregnancies
UN WPP	Crude birth rate × population → Pregnancies, live births; Under-1 pop → DPT, measles	Applies mortality rates for measles denominators	—

Automatic denominator selection

For each indicator, the module selects the denominator that produces coverage closest to the survey benchmark.

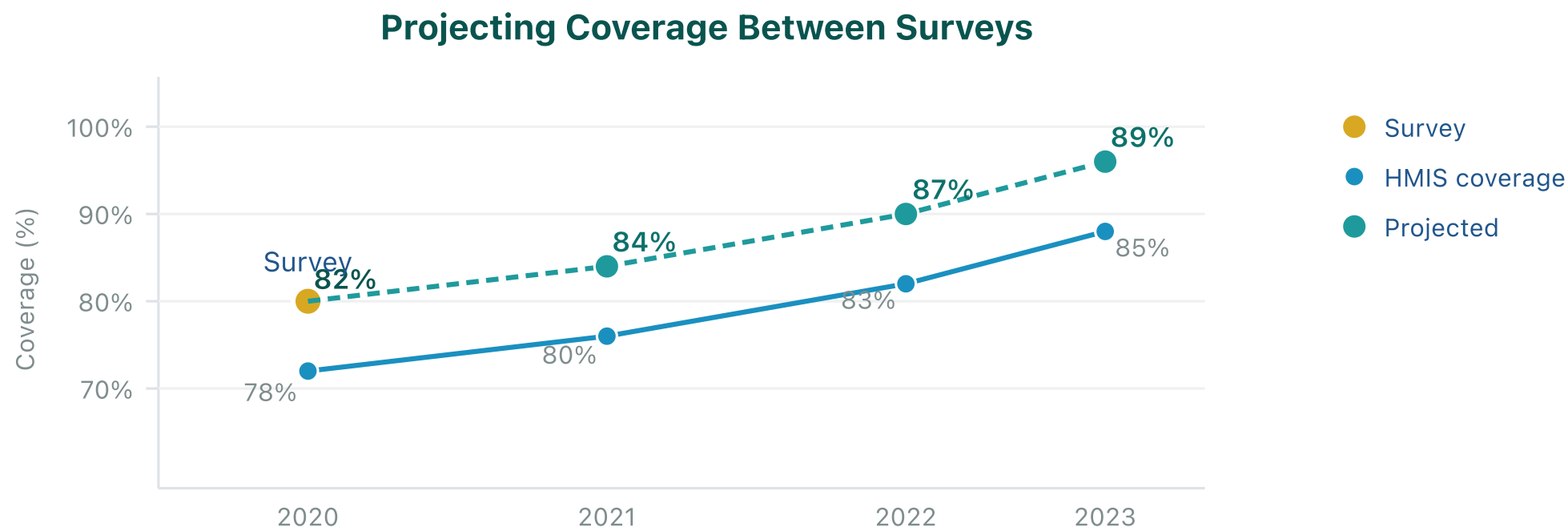
Selection algorithm:

1. Calculate coverage using each denominator option
2. Calculate squared error against survey: $(coverage - survey)^2$
3. Apply selection hierarchy (HMIS-based denominators prioritized over UN WPP)
4. Select the HMIS-based denominator with minimum error

Selection is made per indicator and geographic area. Users may override automatic selections in Part 2.

Coverage projection methodology

The module projects the most recent survey value forward using trends observed in HMIS-derived coverage:



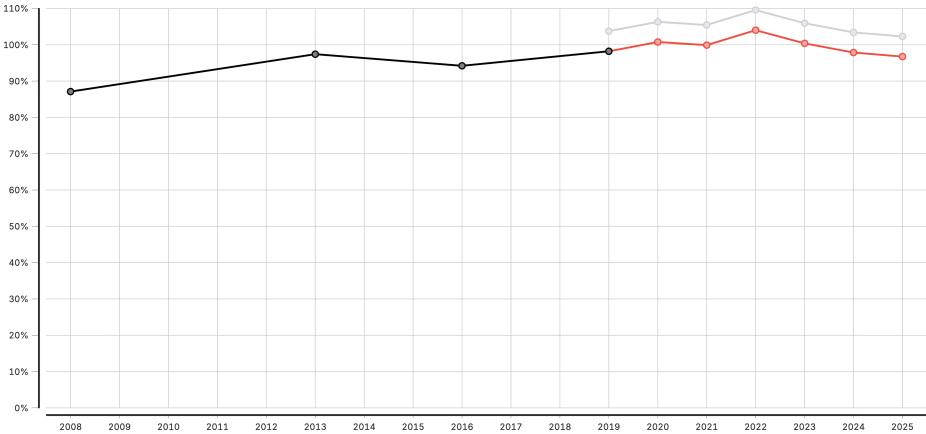
Year-over-year changes (deltas) in HMIS coverage are calculated and applied to the last survey value. This approach preserves the survey baseline while incorporating observed service delivery trends.

Coverage (national)

Coverage estimates for Antenatal client 1st visit

2008 to 2025

DISCLAIMER: These results use routine data to provide rigorous, but not official estimates. They should be interpreted considering any data quality or representation limitations, including data quality findings and any other country specific factors.



Coverage calculated from HMIS data
Projected survey estimate (when survey data is missing)
Survey-based estimate (when available)

Estimating service coverage from administrative data can provide more timely information on coverage trends, or highlight data quality concerns. Numerators are the volumes reported in HMIS, adjusted for data quality. Denominators are selected from UN projections, survey estimates, or derived from HMIS volume for related indicators. National projections are made by applying HMIS trends to the most recent survey data. Subnational estimates are more sensitive to poor data quality, and projections from surveys are not calculated.

MICS data courtesy of UNICEF, Multiple Indicator Cluster Surveys (various rounds) New York City, New York.

DHS data courtesy of ICF, Demographic and Health Surveys (various rounds), Rockville, Maryland.

Data for the current year reflect the period available at the time of analysis; population figures have been scaled to match the corresponding duration.

Interpretation

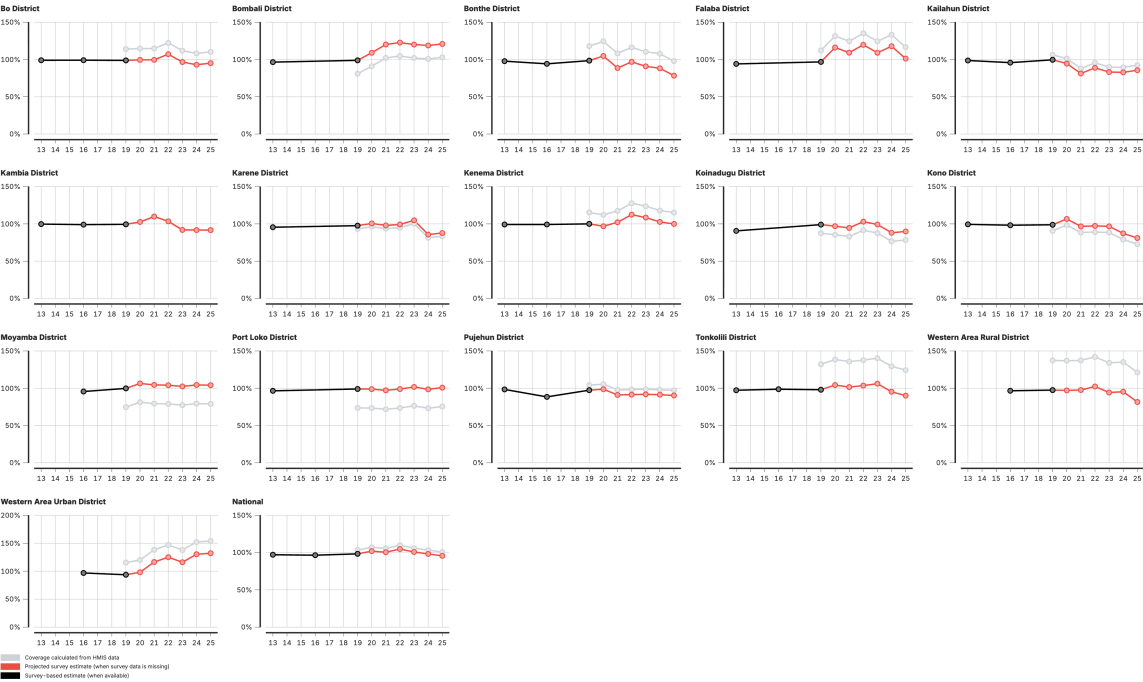
Black line/points show survey data (DHS/MICS) as the household survey reference. **Grey line/points** show HMIS-based coverage from facility data. **Red line/points** show projected coverage — survey estimates extended using HMIS trends.

Coverage (subnational)

Subnational coverage estimates for Antenatal client 1st visit

2013 to 2025

DISCLAIMER: These results use routine data to provide rigorous, but not official estimates. They should be interpreted considering any data quality or representation limitations, including data quality findings and any other country specific factors.



Estimating service coverage from administrative data can provide more timely information on coverage trends, or highlight data quality concerns. Numerators are the volumes reported in HMIS, adjusted for data quality. Denominators are selected from UH projections, survey estimates, or derived from HMIS volume for related indicators. National projections are made by applying HMIS trends to the most recent survey data. Subnational estimates are more sensitive to poor data quality, and projections from surveys are not calculated.

MICS data courtesy of UNICEF, Multiple Indicator Cluster Surveys (various rounds) New York City, New York.

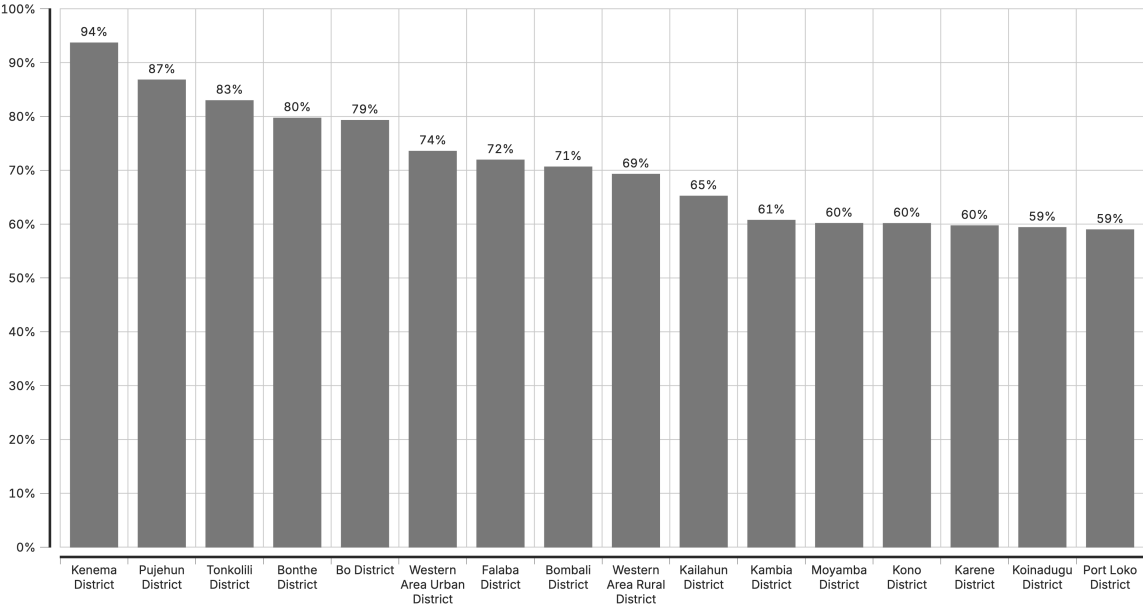
DHS data courtesy of ICF, Demographic and Health Surveys (various rounds), Rockville, Maryland.

Data for the current year reflect the period available at the time of analysis; population figures have been scaled to match the corresponding duration.

Coverage (subnational)

Sub-national level coverage estimates, Antenatal client 4th visit

2025



Estimating service coverage from administrative data can provide more timely information on coverage trends, or highlight data quality concerns. Numerators are the volumes reported in HMIS, adjusted for data quality. Denominators are selected from UN projections, survey estimates, or derived from HMIS volume for related indicators. National projections are made by applying HMIS trends to the most recent survey data. Subnational estimates are more sensitive to poor data quality, and projections from surveys are not calculated.

MICS data courtesy of UNICEF. Multiple Indicator Cluster Surveys (various rounds) New York City, New York.

DHS data courtesy of ICF. Demographic and Health Surveys (various rounds). Rockville, Maryland.

Data for the current year reflect the period available at the time of analysis; population figures have been scaled to match the corresponding duration.

Coverage module: Configuration parameters

Parameter	Description
Count value to use	Which adjusted count to use for coverage calculation
Level to calculate coverage for	Geographic levels for coverage estimation: national, provincial (admin area 2), or district (admin area 3)
Pregnancy loss rate	Proportion of pregnancies ending in loss before delivery
Twin rate	Proportion of deliveries resulting in twins
Stillbirth rate	Proportion of births that are stillbirths
Neonatal mortality rate	Deaths in first 28 days per live birth
Postneonatal mortality rate	Deaths from 28 days to 1 year per live birth
Infant mortality rate	Deaths before age 1 per live birth
Under 5 mortality rate	Deaths before age 5 per live birth

Country-specific mortality rates may be obtained from DHS reports, UN IGME, or national vital statistics.

Session 7: Creating a project

Activity: Creating a Project



In this hands-on session, we will:

- Set up a new project
- Configure project settings
- Select indicators and time periods
- Apply best practices for project organization

Participants will create their first project

Session 8: Creating visualizations

Activity: Creating Visualizations



In this hands-on session, we will:

- Explore available chart types
- Create and customize visualizations
- Export charts for use in reports

Participants will build visualizations from their analysis



Lunch Break

90 minutes

Back at 14:00

Session 9: Creating reports

Activity: Creating Reports



In this hands-on session, we will:

- Use report templates
- Generate automated reports
- Customize report content and layout

Participants will create their first quarterly report draft

Day 3

Recap

On Day 2, we covered:

- FASTR methods for data quality assessment and adjustment
- Creating and configuring projects in the platform
- Building visualizations from Zambia data
- Generating reports

Day 3 Focus

Today we move from analysis to action:

- Deep dive into interpretation of results
- Create the Q4 2025 quarterly report
- Present findings to the group
- Develop action plans for continued work

Agenda

Day 3 -- From Analysis to Action: Interpreting Results and Using FASTR for Decision-Making

Time	Agenda	Facilitator/Presenter
09:00–09:15	Recap and Day 3 agenda	GFF FASTR team
Session 10: Interpretation of visualizations		
09:15–10:15	Approaches to support interpretation	GFF FASTR team
Session 11: Creating a Q4 2025 report		
10:15–12:30	Creating short and long reports with country context	GFF FASTR team
12:30–14:00	<i>Lunch Break</i>	
Session 11 (cont'd): Creating a Q4 2025 report		
14:00–15:00	Continue report creation with country context	GFF FASTR team
Session 12: Presenting reports		
14:30–15:30	Present reports, group feedback	GFF FASTR team

Session 10: Interpretation of visualizations

Analytical thinking & interpretation

Interpretation connects **data patterns** to **programmatic meaning**.

For every FASTR output, ask three questions:

- 1. What does it show?** — Describe the pattern accurately
- 2. Why might that be?** — Consider multiple explanations
- 3. What should we do?** — Identify next steps or actions

Moving from numbers to insights requires context, critical thinking, and programmatic knowledge.

Critical thinking checklist

Before drawing conclusions from FASTR outputs, ask:

- **Data quality or real change?** Could this pattern reflect reporting issues rather than service delivery changes?
- **What external factors apply?** Policy changes, seasonal patterns, shocks, or campaigns?
- **Does it make programmatic sense?** Would this pattern be expected given what we know about the health system?
- **What does disaggregation reveal?** National trends can mask important regional variation
- **What additional information is needed?** What context would strengthen or challenge this interpretation?

Common interpretation pitfalls

Aggregation masking

National averages can hide critical regional variation. Always examine subnational patterns.

Completeness $\neq$ representativeness

High reporting completeness doesn't mean data captures all service delivery (private sector, informal care).

Missing context

Patterns without context are meaningless. Consider seasonality, policy changes, and external shocks.

Over-interpreting noise

Short-term fluctuations may be normal variation. Focus on sustained trends and significant deviations.

Interpreting FASTR module outputs

Data quality assessment

Low DQA scores may reflect data entry practices rather than service quality. Focus on identifying the specific dimension (completeness, outliers, consistency) driving issues.

Service utilization

Distinguish seasonal patterns from real disruptions. Year-over-year comparisons help isolate programmatic changes from normal fluctuation.

Coverage estimates

Coverage >100% signals reconciliation needed between HMIS and survey data. Consider whether the denominator source matches the service delivery context.

Across all modules

Examine patterns at multiple geographic levels — subnational disaggregation often reveals the true story.

Session 11: Creating a Q4 2025 report

Generating quarterly reporting products

Content to be developed

This section will cover:

- Quarterly reporting workflow
- Using the FASTR platform for automated reports
- Quality assurance for reports
- Distribution and feedback mechanisms



Lunch Break

90 minutes

Back at 14:00

Session 12: Presenting reports

Presenting reports and group feedback

Content to be developed

Day 4

Recap

On Day 3, we covered:

- Interpretation of FASTR visualizations
- Creation of the Q4 2025 quarterly report
- Presentation of findings and group feedback
- Action planning for continued platform use

Day 4 Focus

Today we design the Health Facility Assessment:

- Overview of the FASTR HFA phone survey
- Review questionnaire structure
- Adapt questionnaire for Zambia context
- Plan HFA priorities and data use

Agenda

Day 4 -- Designing the Health Facility Assessment

Time	Agenda	Facilitator/Presenter
09:00–09:15	Recap and Day 4 agenda	GFF FASTR team
Session 13: Overview of FASTR HFA phone survey		
09:15–10:15	HFA overview and questionnaire adaptation guidelines	GFF FASTR team
Session 14: Questionnaire adaptation to the Zambian context		
10:15–12:30	Review questionnaire + hands-on adaptation	GFF FASTR team
12:30–14:00	<i>Lunch Break</i>	
Session 14 (cont'd): Questionnaire adaptation		
14:00–15:00	Continue questionnaire adaptation (in groups)	GFF FASTR team
Session 15: Discussion: HFA adapted questionnaire		
14:30–15:30	Discuss adapted questionnaire in plenary	GFF FASTR team
Session 16: Discussion: HFA priorities and data use		
15:30–16:30	HFA priorities and data use case in Zambia	GFF FASTR team
Session 17: Action planning and wrap-up		
16:30–17:00	Key messages and wrap-up	GFF FASTR team

Session 13: Overview of FASTR HFA phone survey

Health Facility Assessment Phone Survey

Placeholder: Content to be developed



Lunch Break

90 minutes

Back at 14:00

Next Steps & Action Planning

Key actions to take after this working session:

- Finalize adapted phone survey questionnaire
- Complete first quarterly report draft
- Train remaining MoH team on data downloader tools
- Prepare roadmap for participation in Abuja workshop

Quarterly Reporting Cycle

Establish a sustainable reporting rhythm:

- Define quarterly report timeline and responsibilities
- Set up data extraction schedule
- Establish review and approval process
- Plan dissemination to stakeholders

Continued Platform Usage

Maintain momentum with the FASTR platform:

- Regular data updates and quality checks
- Expanding analysis to additional indicators
- Building capacity across MoH teams
- Documentation of lessons learned

Workshop Closing

Thank you for your participation!

Over the past four days, we have:

- Configured the FASTR platform for Zambia
- Created the first quarterly RMNCAH-N report
- Adapted the HFA questionnaire for local context
- Built capacity for ongoing analysis and reporting

Stay Connected

Continue the journey:

- Platform access and support
- Quarterly report submissions
- Multi-country workshop in Abuja
- Ongoing technical assistance from GFF team

Thank You

FASTR Working Session – Zambia
January 27–30, 2026
Lusaka

Contact Information

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